Review of last week's lecture

- **Preinitiation:**

  tRNA charging (adding the correct amino acid to the correct tRNA); Dissociation of ribosomes

- **Initiation:**

  assembly mRNA, ribosomes, and tRNA

- **Elongation:**

  reading the code and converting the information into a peptide

- **Termination:**

  ending the synthesis of a protein

- **Post-translational modification**

Functional Genomics

Use of ‘genomic scale” technologies to understand the biological functions of genes and DNA sequences.

截至 2026 年 5 月，全球已完成基因组测序的物种总量超过 18 万种；

其中真核生物约 3.7 万种，

原核生物（细菌 / 古菌）约 14.3 万种。

NCBI RefSeq

Approaches

- Gene-by-gene assessment of gene function
- Genetic dissection of biological pathways
- Genetic identification of regulatory elements
- Tool development
  - Specialized libraries (like yeast barcoded KOs)
  - Collections of reagents to disrupt genes
  - Genetic reagents
    - engineered chromosomes
    - Special stocks/strains of organisms
    - P-element collections, for example
  - New experimental/computational methodologies

Genome Project Resourses

- **Sequences of various genomes**
- **Sequences of most expressed genes**
- **Gene expression patterns (RNA sequencing, microarrays etc.)**
- **Proteomics；　Metabolomics；　Phenomics**
- **Gene functions?**

Choosing tools for job

- Basic cellular processes
  - Yeast; cultured cells
- Cell fate determination
  - *C. elegans*
- Fundamental body plan
  - *Drosophila*
- Vertebrate development
  - Zebrafish
- Disease models/ human relatedness
  - Mice
- Gene regulation
  - Informatics; comparative genomics; molecular biology; eQTL …

Functional analysis of the *Caenorhabditis elegans* genome using RNAi

- Screened 86% of 19,427 genes
- Cataloged each gene based on phenotype
- 1,722 produced a mutant phenotype
- 2/3 of the mutant phenotypes were associated with previously uncharacterized genes
- RNAi is not complete "knockout," rather "knock down."

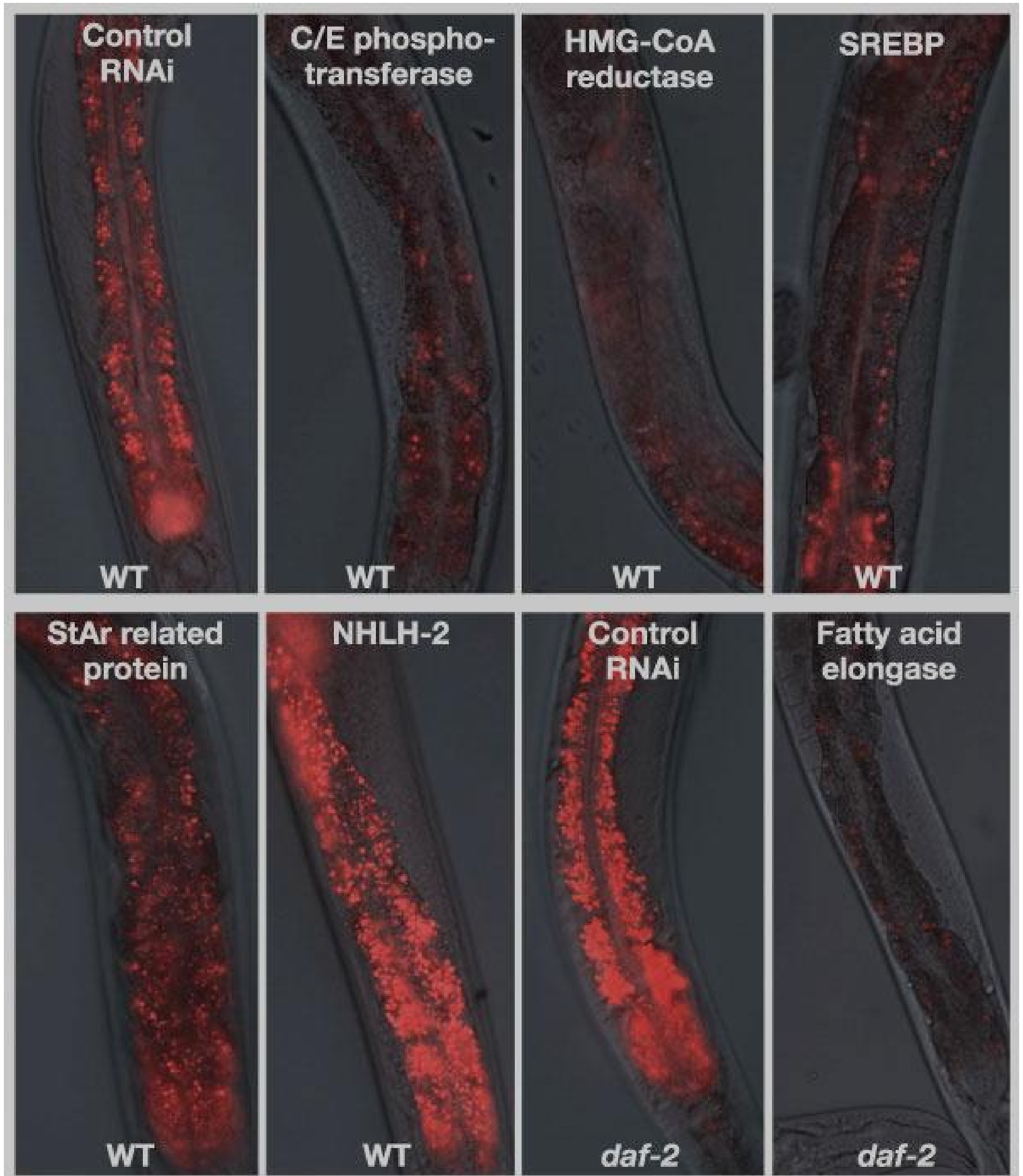
• Embryonic lethality
• Sterility
• Slow growth
• Larval arrest
• Adult lethal
• High incidence males
• Multivulva
• Paralysis
• "dumpy"
• Etc.

Kamath et al *Nature* **421**, 231-237 (2003)

Key findings

- Phylogenetically conserved genes most likely to be affected
- Gene duplicates are often partially or fully redundant
- Genes on X-chromosome are not required during embryonic or larval development, but do have important roles in the adult

Genes affecting fat content in worms



RNAi screen of worms eating *E. coli* grown in Nile Red (binds fat storage droplets).

- Metabolic enzymes
- Transcription factors
- Fatty acid elongation enzymes

Conclusions

- Out of 16,757 genes tested:
  - 305 reduced fat or distorted fat deposition pattern
  - 112 increased fat or enlarged fat droplet size
  - 261 reduced fat, but also other complications
- Nervous system involvement

Two strategies in functional genomics

Investigate the function of a special gene

Phenotype driven approach：Forward genetic strategy

Gene driven approach: Reverse genetic strategy

Forward Genetics

Phenotype $\longrightarrow$ Gene

Mutations: A Geneticist's Best Friend

- **Defects in genes give clues to function**
- **Different classes of mutations:**
  - Dominant
  - Semi-dominant
  - Recessive
  - Different "morphs" (hypomorphs, hypermorphs, antimorphs). Widely used in Drosophila.
- **How to get mutations: spontaneous vs induced**

Spontaneous mutants - you get what you see

Piebald(花斑)

Albino, chinchilla 南美栗鼠

Hairless, rhino allele

Agouti mutations

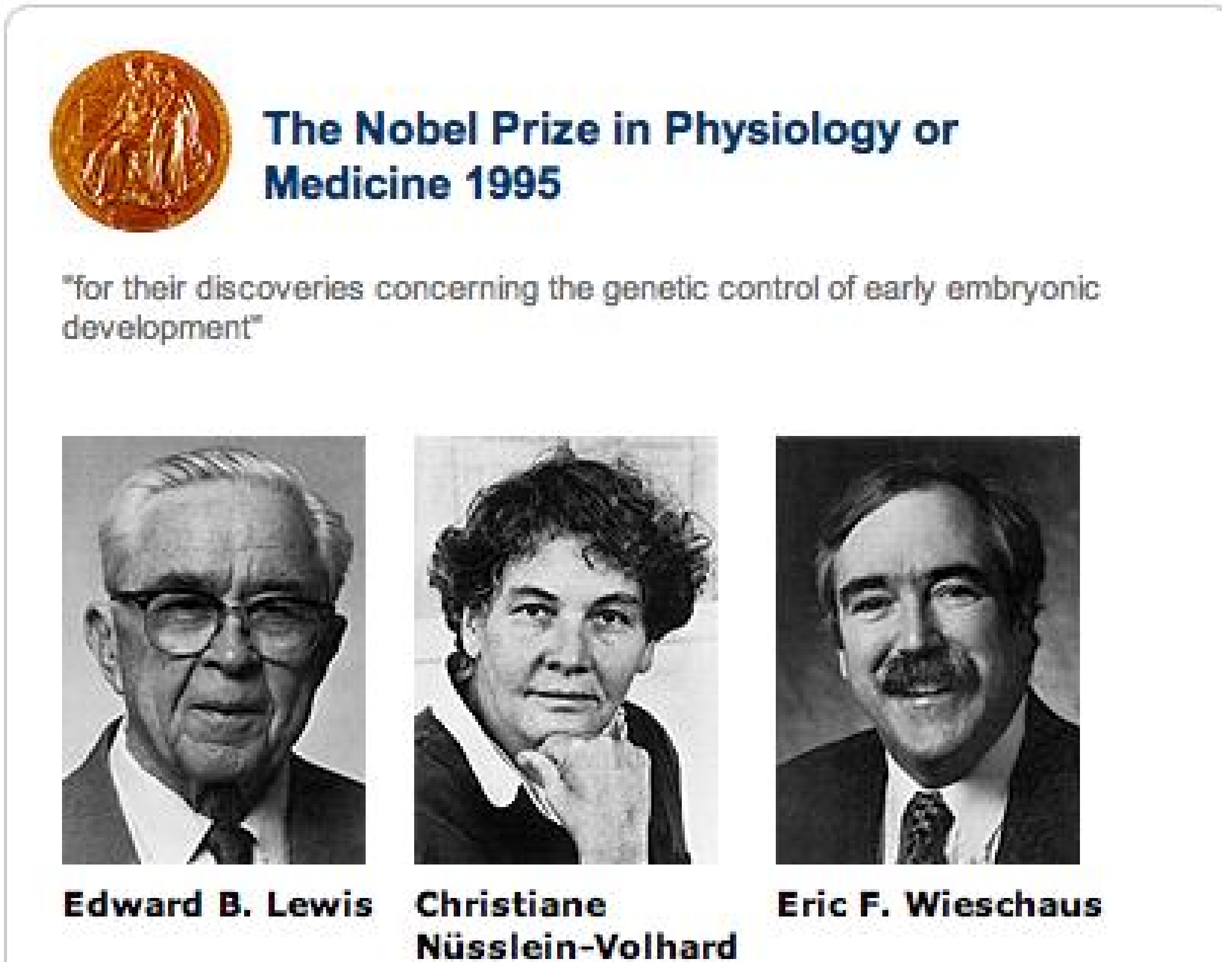
belted

A/A; brown

Why induced mutagenesis?

- Technical Reasons:
  - Spontaneous mutations too rare
  - Futile to screen for subtle, spontaneous phenotypes
- Biological Reasons:
  - Find genes involved in certain pathways
  - To understand what all genes do

Nobel Prize for Forward Genetics



刘易斯、福尔哈德、威斯乔斯

他们的研究揭开了胚胎如何由一个细胞发育成完美的特化器官,如脑和腿的遗传秘密，也建立了动物基因控制早期胚胎发育的模式

Genetic “Screens”

- Treat organism with mutagen that affects germline
- Outcross (diploids) to establish lineages or find dominant mutations
- Intercross to detect recessive mutations
- Screen progeny for mutant phenotypes

Types of forward genetic screens

- Mutagenic Agents:
  - Physical agents (X-ray， γ-ray)
    - Induces deletions
    - Complicates genetic analysis
  - Chemical “point” mutagens (ENU; EMS)
    - Highly potent; many germline mutations
    - Many alleles and “morphs”
    - Difficult to identify mutation
  - Insertional mutagens
    - Fewer mutations/genome
    - Easy to clone

Types of forward genetic screens II

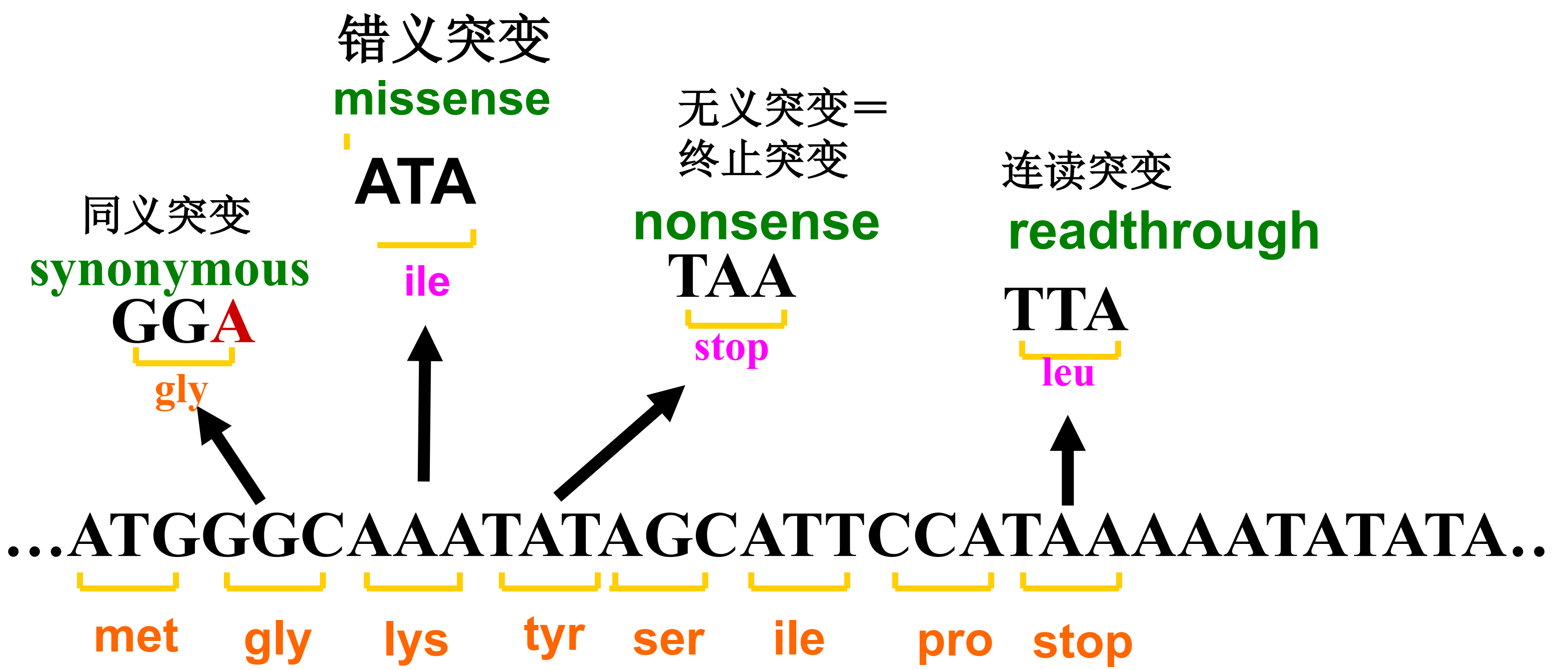
- Genome-wide
  - Casts wider net
  - But chromosome location unknown, so Need to genetically map
- Region-specific
  - Need special genetic reagents (insertions; deletions; inversions)
  - Know the location of mutation
  - Only access part of genome

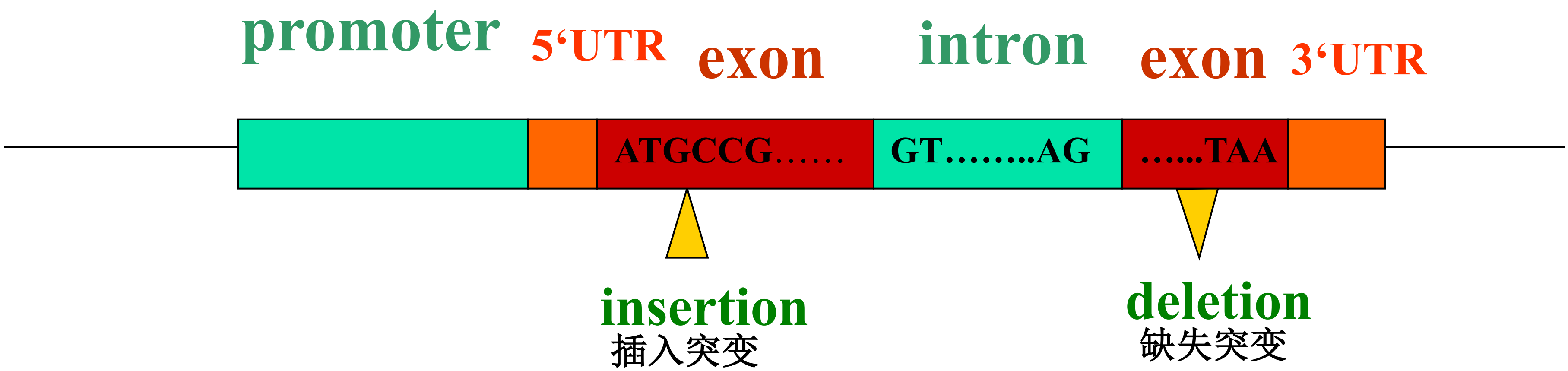
Chemical mutagenesis

- Ethylnitrosourea (ENU) 乙酰基亚硝基脲
  - Zebrafish, mice
- Ethylmethanesulfonate (EMS)甲磺酸乙酯
  - Drosophila; C.elegans; Arabidopsis

Key: they need to mutate the germ cells at high efficiency

The effect of point mutation on protein





Replication slippage(复制滑移): In DNA replication when parental strand has short repeat sequence, the slippage will occasionally occur between parental strand and it's complementary daughter strand, that cause the repeat unit sequence replicate several times or lose several times.

Microsatellite markers in map based cloning

Huntington disease

Gain-of –function:
gain abnormal
protein acitivity

Ectopic expression, for example activation tagging

Increased expression in specific tissue, for example enhancer trap

Mutation occur in destructive domain (responsible for degradation by ubiquitin system), which cause protein abnormally stable, for example Aux/IAA family protein in plant

Insertional mutagenesis

- Insert vector into genome that disrupts genes
- Allow it to happen randomly
- Make collection of individuals, each with different insertion(s)
- Do crosses to make insertion homozygous
- Screen for phenotypes
- Use inserted DNA to identify mutated gene

Insertional mutagens

T-DNA insertion (plants)

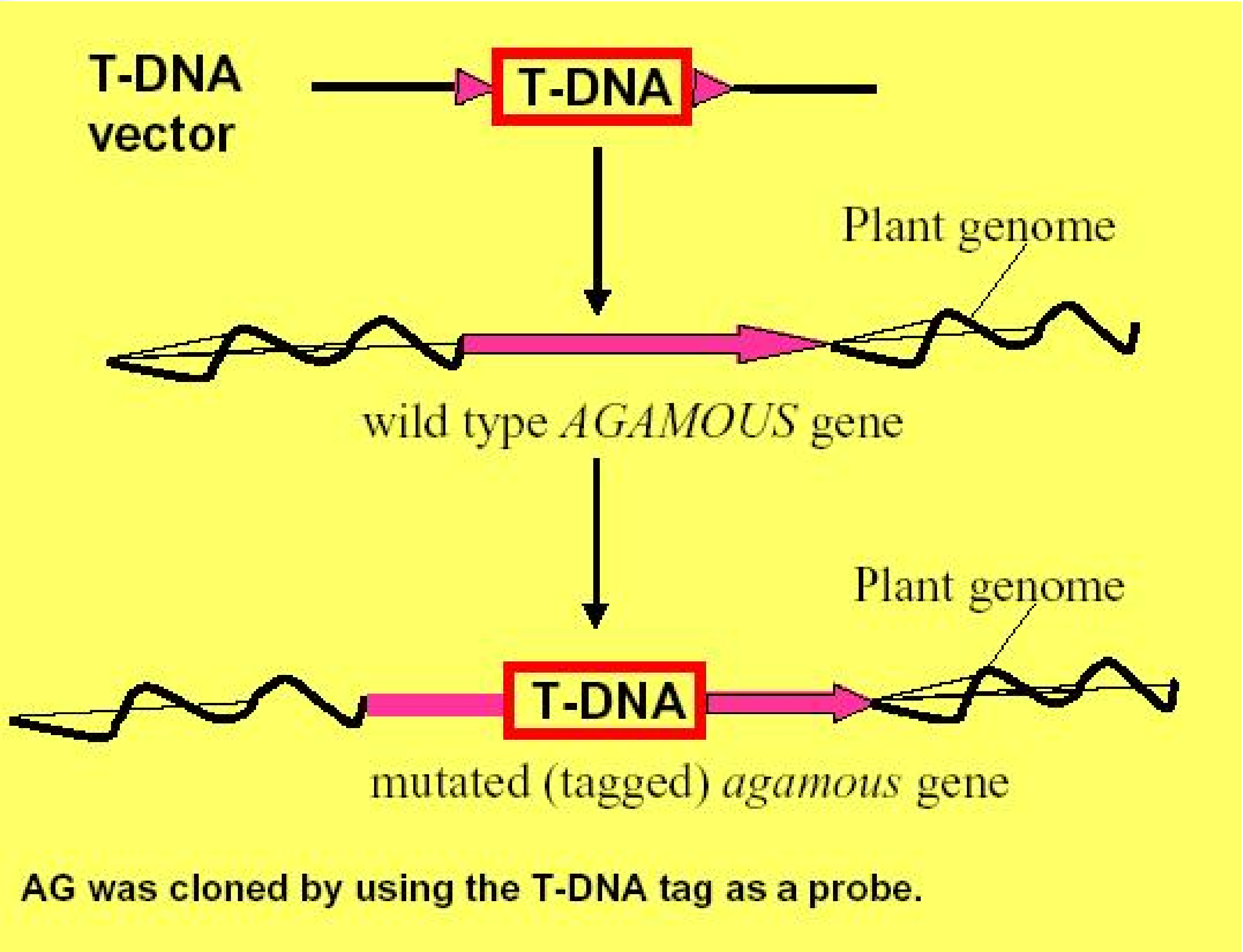
Transposable elements

Mobile elements jump from introduced DNA
P elements in *Drosophila*

Mobilize by introducing active element
AC/DS elements in plants

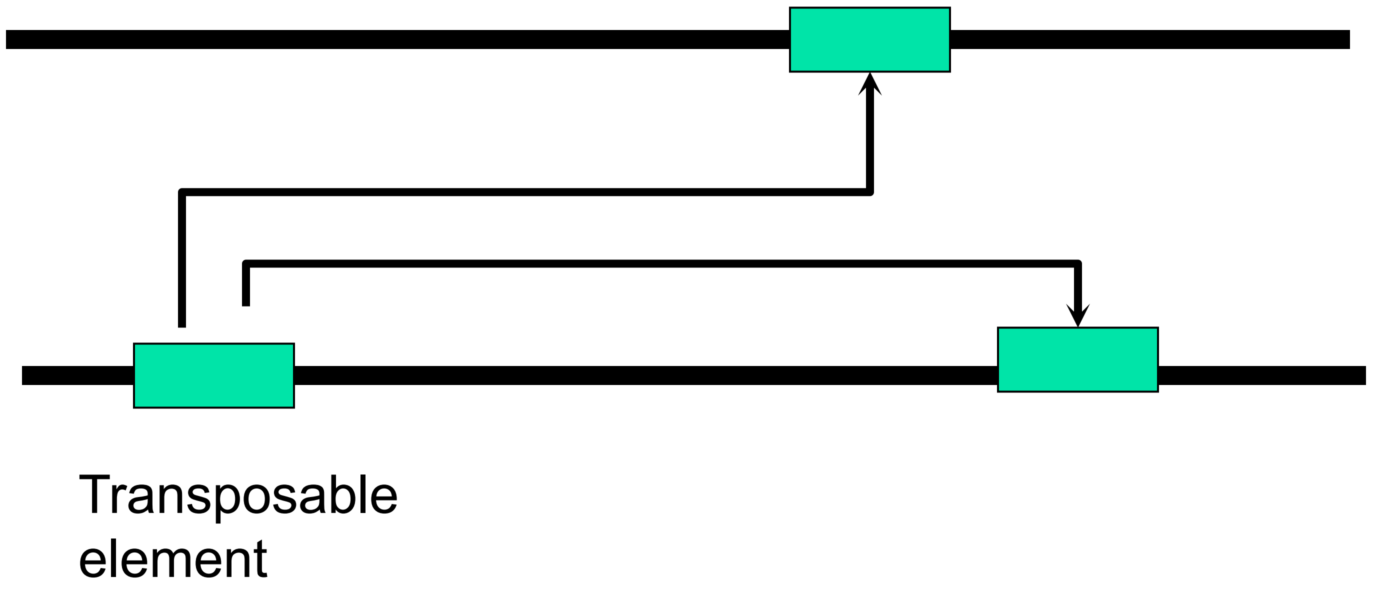
Infect/transfect with retroviral insertions or other vectors
popular for zebrafish
also in plants

T-DNA tagging mutagenesis



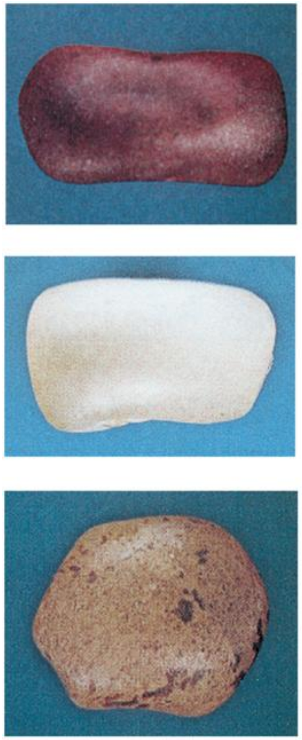
Transposable elements

- Mobile genetic elements - they move from one location in the genome to another



The Nobel Prize in Physiology or Medicine 1983

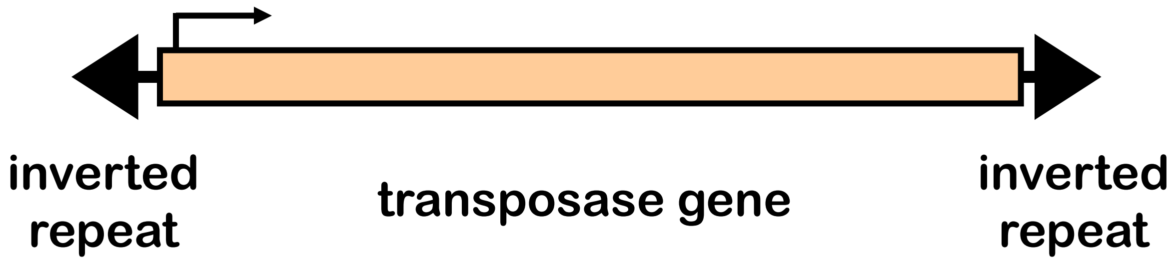
Discovered by Barbara McClintock



Observations of B. McClintock (1930's-1950's)

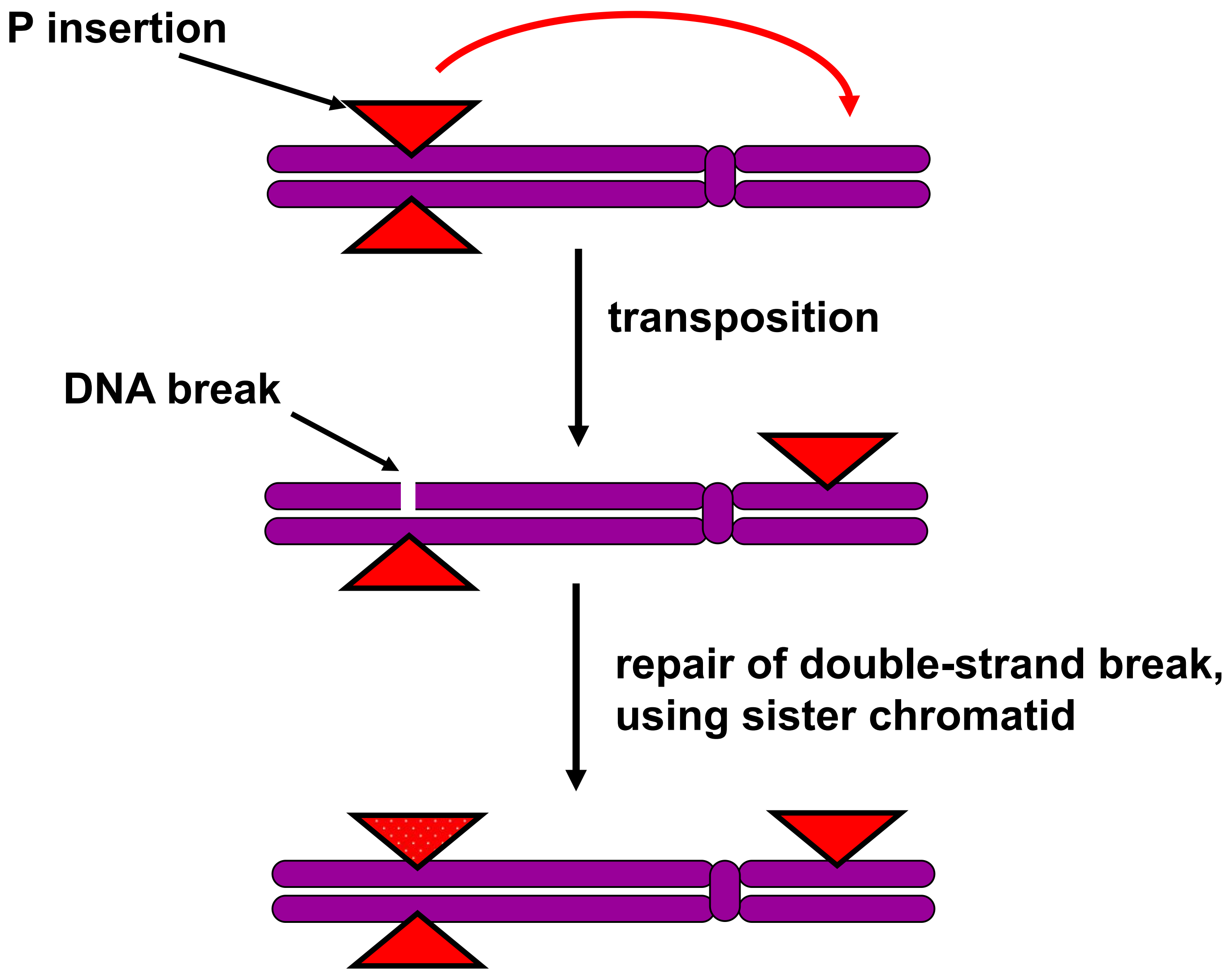
Certain crosses in maize resulted in large numbers of *mutable* loci.
– The frequency of change at those loci is much higher than normally observed.
• Studies of these plants revealed a genetic element called "Dissociation" or *Ds* on the short arm of chromosome 9.
• Chromosome breaks occurred at the *Ds* locus, which could be observed cytologically
– i.e. by looking at chromosome spreads from individual cells, e.g. sporocytes.
• Frequency and timing of these breaks is controlled by another locus, called "Activator" or *Ac*.

Insertional Mutagenesis with Transposable Elements

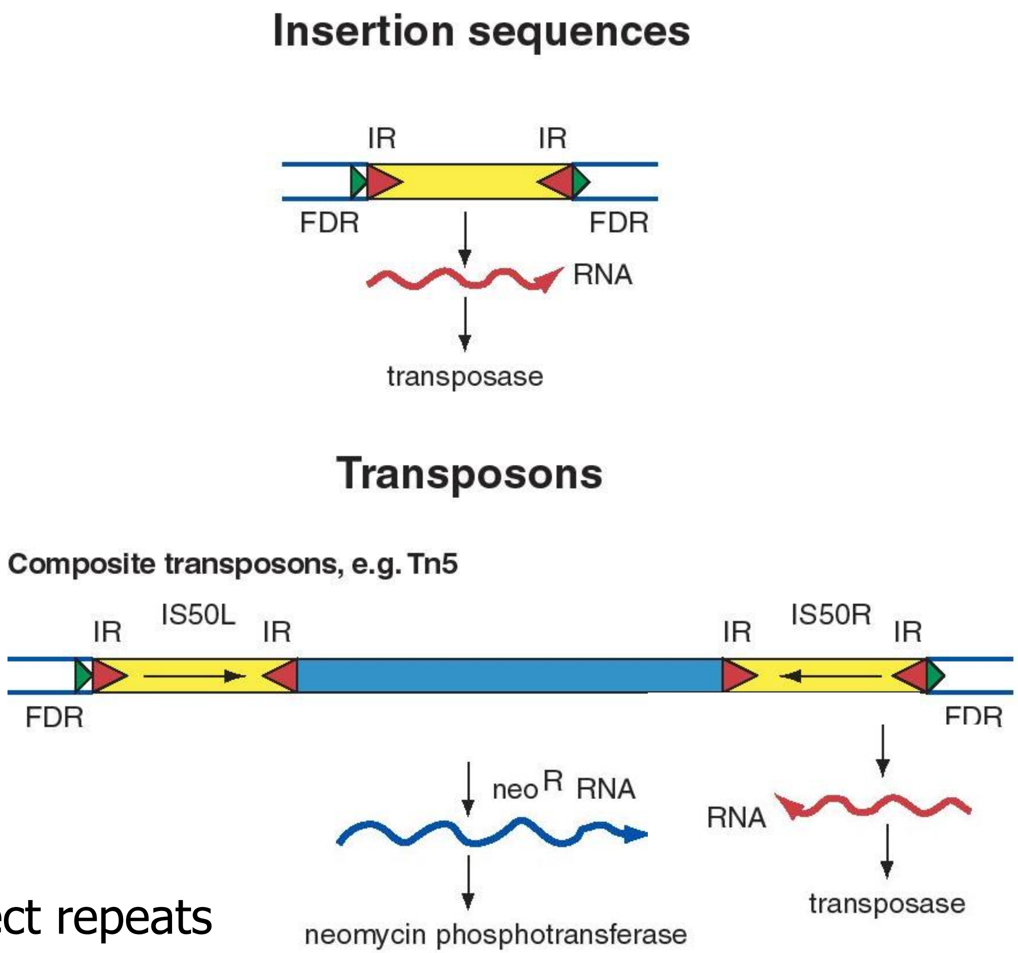


Drosophila P element

Insertional Mutagenesis with Transposable Elements



IS elements and transposons



FDR, Flanking direct repeats

Transposon mutagenesis

Maize transposon systems:

• *Mu*: *MuDR* (autonomous), *Mu1* to *Mu8* (non-autonomous； only maize)

• *Enhancer/Supressor-mutator* (*En/Spm*): *En* or *Spm* (autonomous), *I* or *dSpm* (non-autonomous)

• *Activator/Dissociation* (*Ac/Ds*): *Ac* (autonomous), *Ds* (non-autonomous)

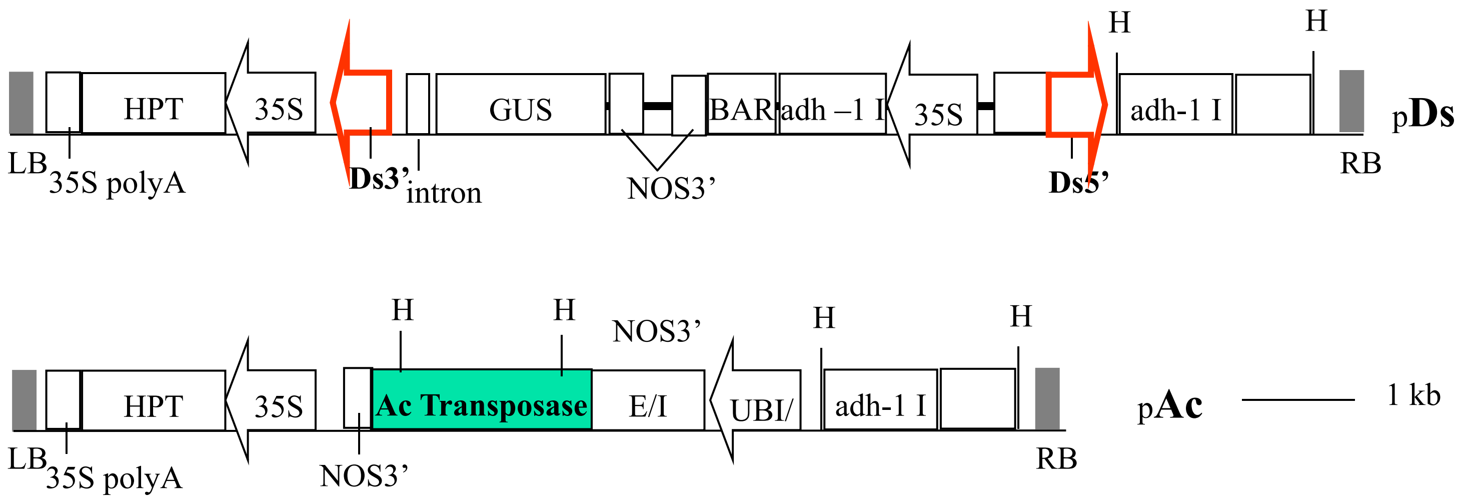
Development of transposon systems for other plant species:

• T-DNA with transposable elements

• Transformation with non-autonomous element

• Cross with plant transformed with immobilised autonomous element ($\rightarrow$ transposition)

T-DNA with transposable elements and non-autonomous element



Schematic representation of the T-DNA containing the transposon (Ds) (up) and AcTPase (lower), as well as selectable markers associated with each. Both vector are based on pCAMBIA1300. Abbreviations: LB, T-DNA left border; RB, T-DNA right border; HPT, hygromycin phosphortransferase; 35S, CaMV35S promoter; adh-1 I, maize adh-1intron; nos3', 3' signal of nopaline synthase; GUS, β-glucuronidase; 35S polyA, CaMV35S polyA signal; Ds5', 5' termini of dissociation from maize, Ds3', 3' termini of dissociation from maize, Δ35S, CaMV 35S mini promoter; intron, intron of gene for G-protein α-subunit from Arabidopsis; UBI/E/I, maize ubi-1 promoter/Exon/Intron; Ac transposase, transposase of activator from maize.

Transposon mutagenesis

Ac start line × Ds start line

Plant with immobilized autonomous element

Plant with non-autonomous element

↓

F1 transposition

↓

F2 Select on PPT^R lines

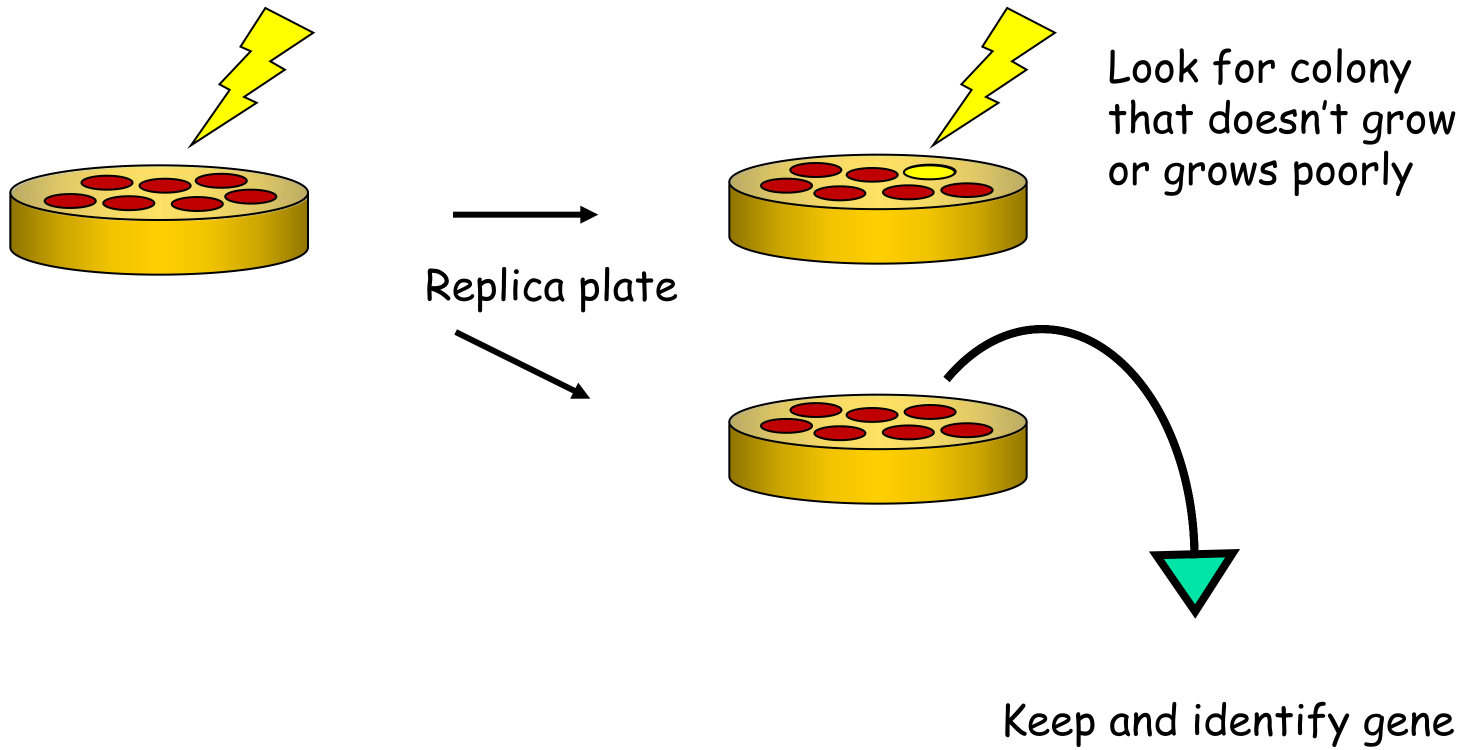
PCR identification of transposition. Selection for plants with stable insertion into other gene and without transposase

Mutagenesis of yeast

Example: isolation of mutants required for DNA repair

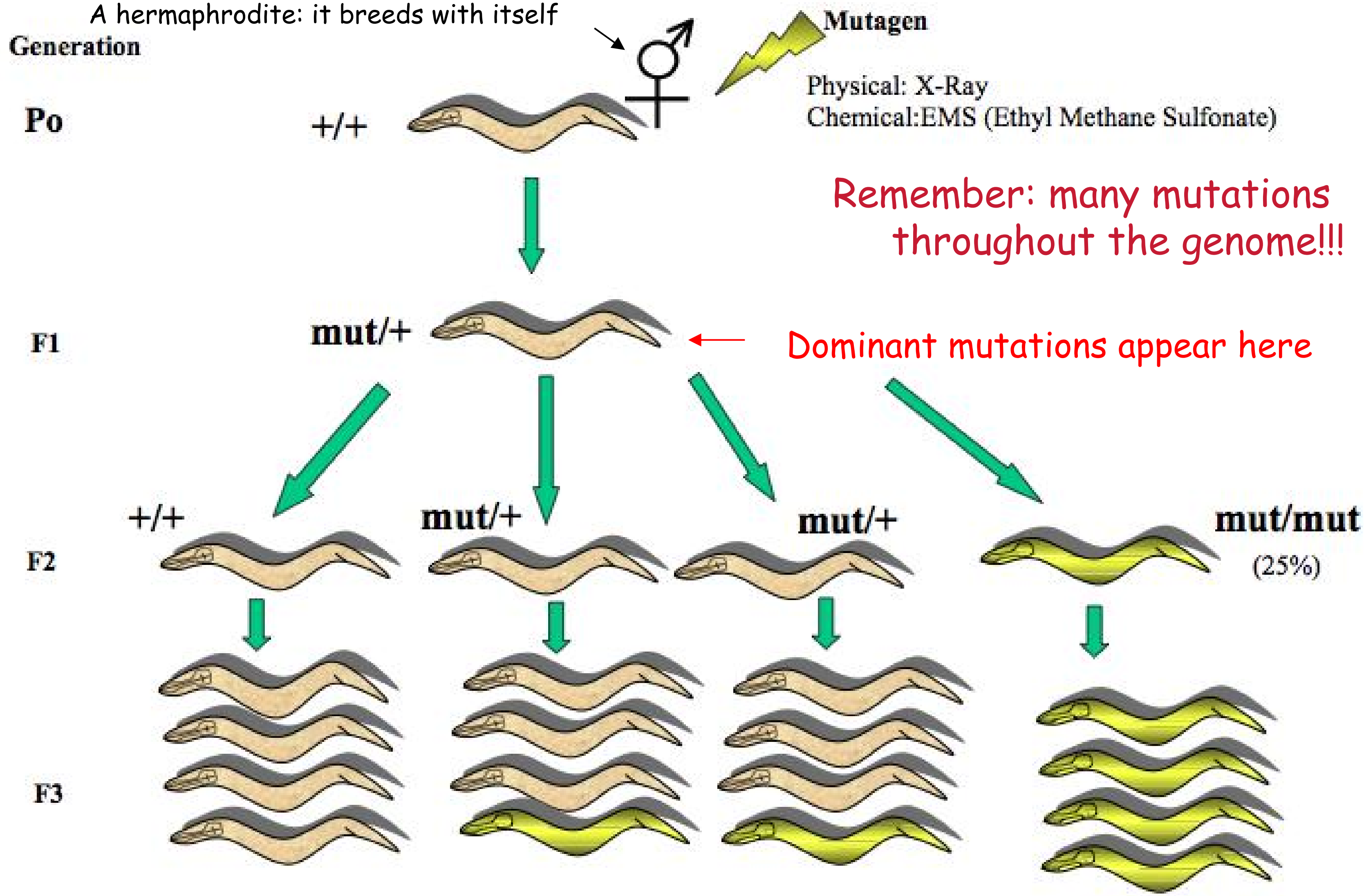
Mutagenized with MMS(甲基汞化硫酸酯)

Radiation **(or other DNA-damaging agents)**



Genome-wide screens in *C. elegans*

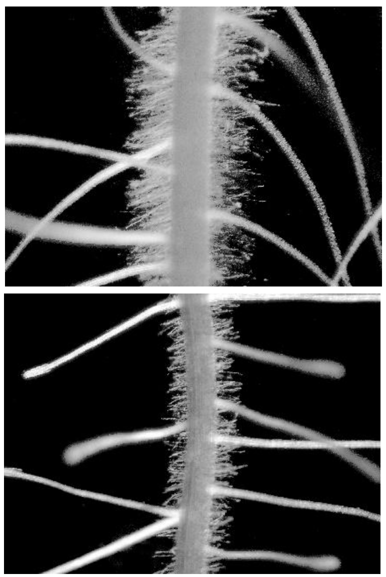
A hermaphrodite: it breeds with itself



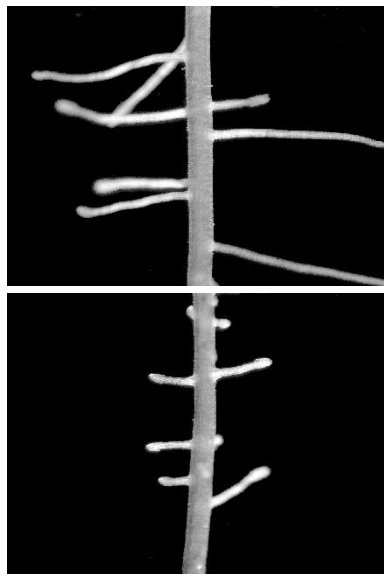
Remember: many mutations throughout the genome!!!

Dominant mutations appear here

Mutant screen in our lab

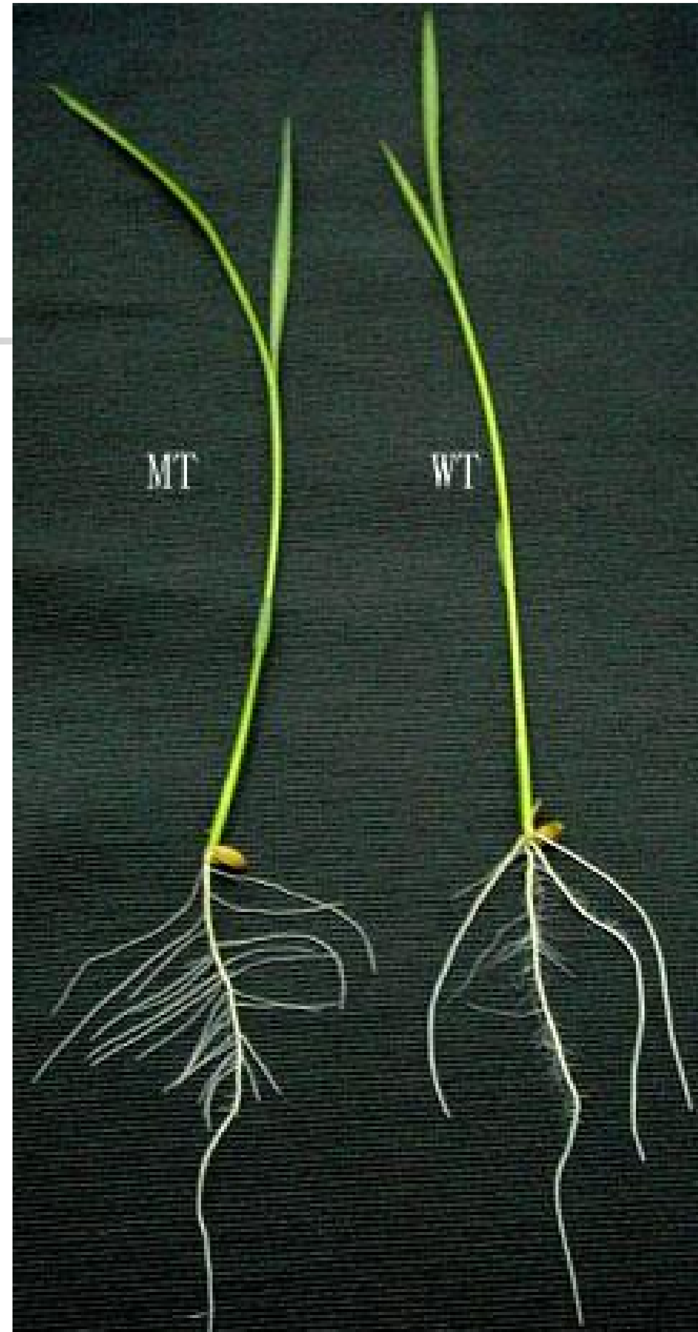


WT(HJ)



hairless

10d seedlings



Crown rootless mutant （left）

Types of alleles

Recessive

Amorphs: nulls

Hypomorphs: partial loss-of-function

Dominant

Antimorphs: counteracts wt

- dominant negative

Hypermorphs: gene is “overactive” or more stable

NeoMorphs: gene gets “new” function

-expressed in new location

3 methods to clone the insertion site

Inverse PCR

TAIL-PCR

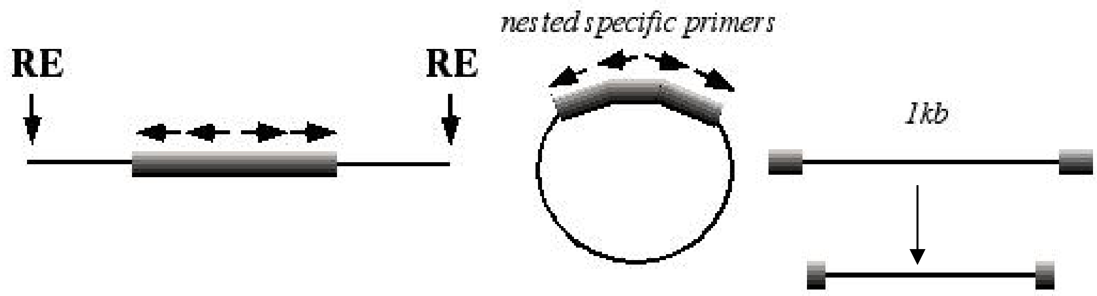
Plasmid rescue

Apply to

T-DNA insertion

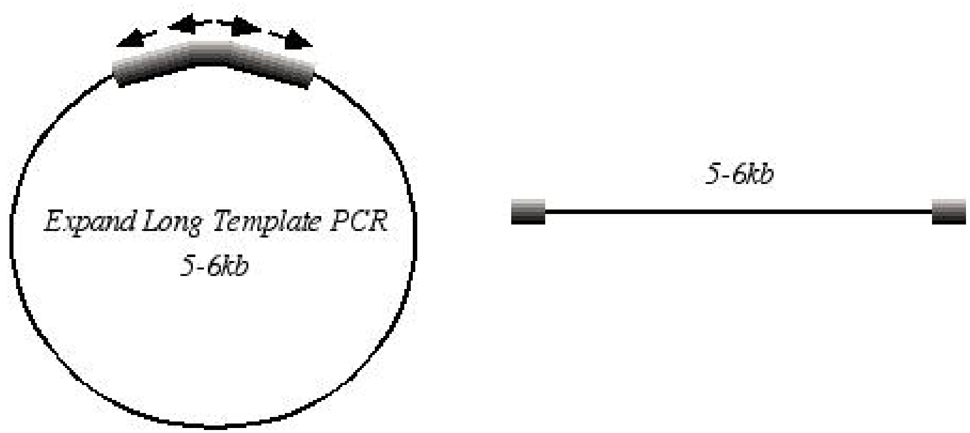
Transposable elements insertion

Isolation of flanking regions by inverse PCR (IPCR)

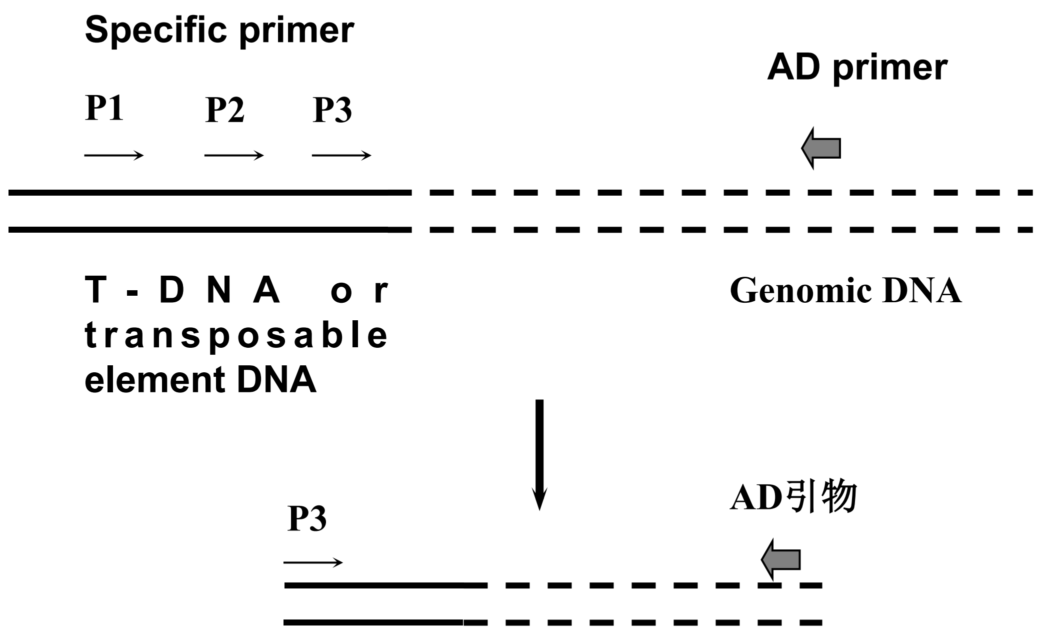


LR-IPCR = Long Range IPCR

(Mathur et al., 1998)



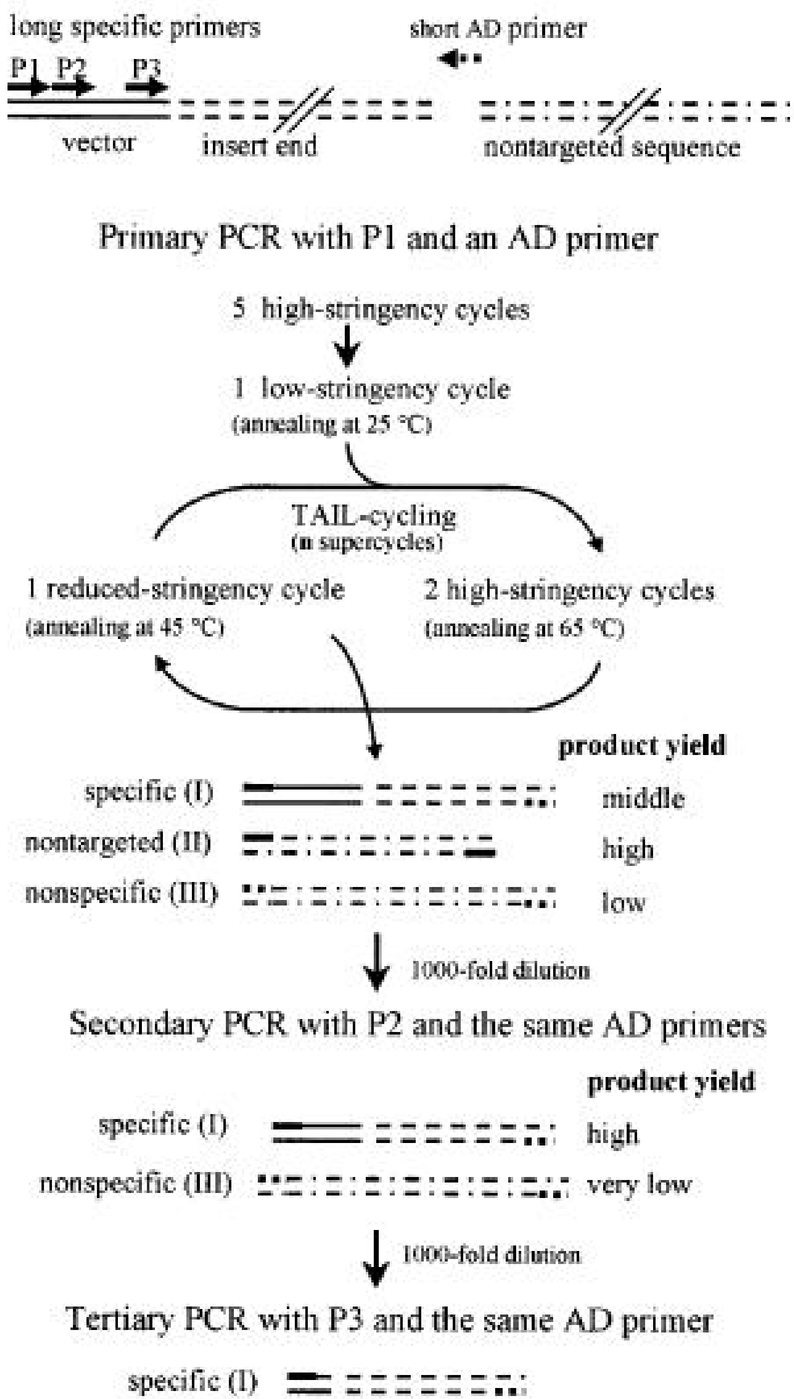
TAIL-PCR principle



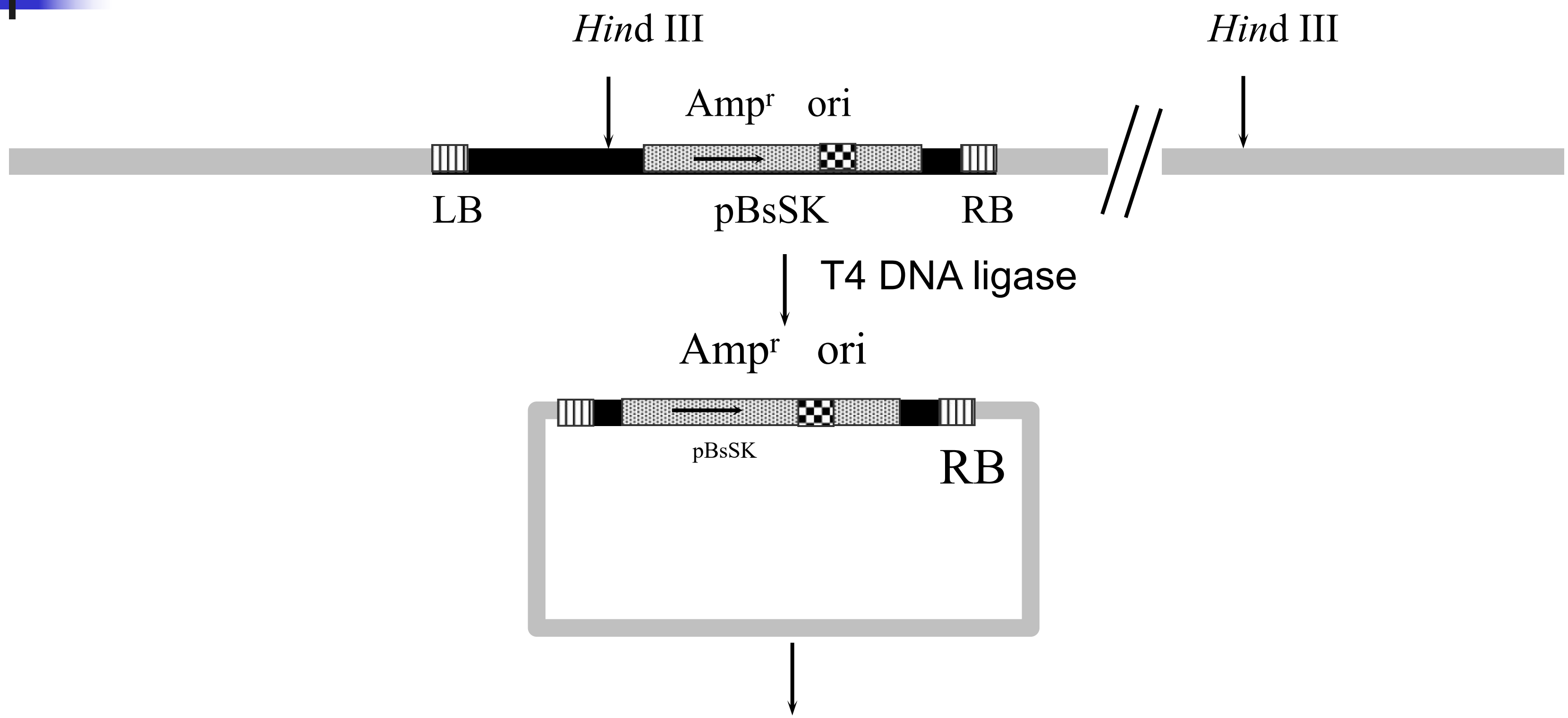
Schematic representation of amplification of genome sequence flanking insertion site with TAIL-PCR.

TAIL-PCR process

Thermal asymmetric interlaced polymerase chain reaction



Plasmid rescue



Transform into E. coli，get Amp^r clone

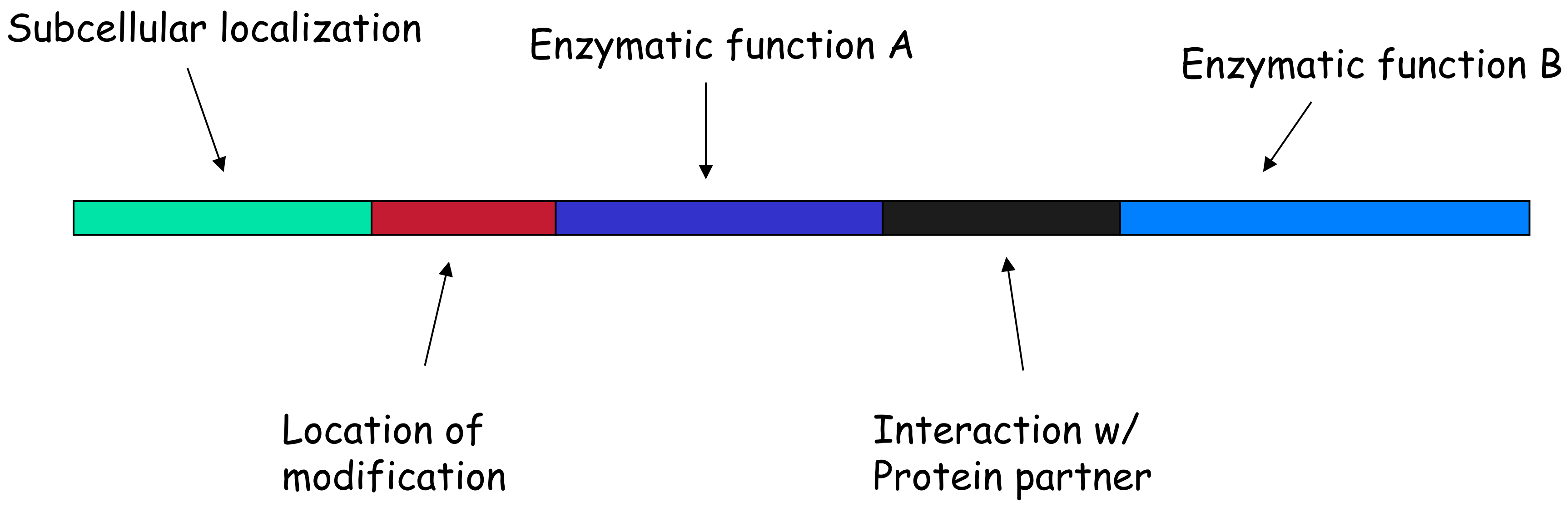
Schematic representation of cloning genome sequence flanking insertion site with plasmid rescue.

How to clone the point mutations?

- That's the drawback of chemical mutagenesis
- Need to do linkage analysis and recombinational mapping "Genetic mapping"

Screens with T-DNA insertional lines simplify the linkage analysis, since you know what chromosome the mutation is on.

Alleles and functional insight



- Diverse alleles are best obtained with point mutations

Map based cloning

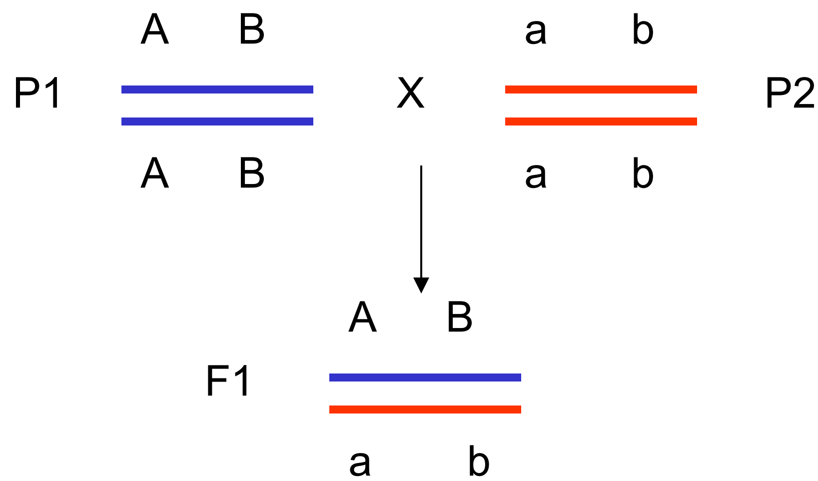
How to clone mutated target gene by positional cloning?

A powerful tool in forward genetic research---------

Map based cloning: Involves isolating a marker (piece of DNA) whose location is known and which is **tightly linked** to the gene you are trying to clone

linkage map= **genetic map (遗传图谱)**

Linkage analysis: during meiosis, 2 copy homo chromosome will respectively repeat itself and form bivalent. Crossing-over or recombination will occur. From the freq of crossing-over, the linkage distance can be estimated.



AB	Ab	aB	ab
(1-r)/2	r/2	r/2	(1-r)/2
1/2	0	0	1/2
1/4	1/4	1/4	1/4

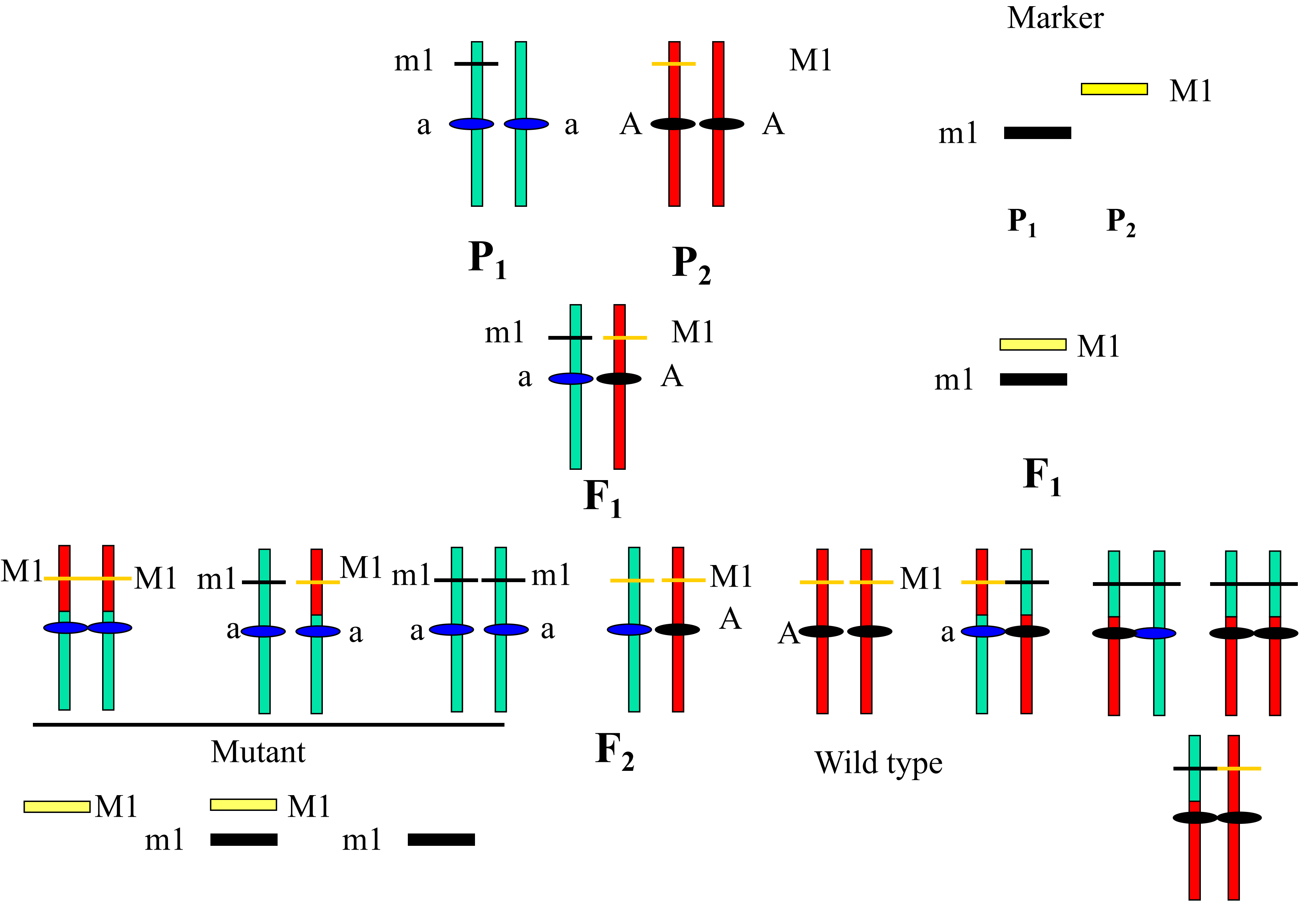
Linkage (0<r<0.5)

Completely linkage (r = 0)

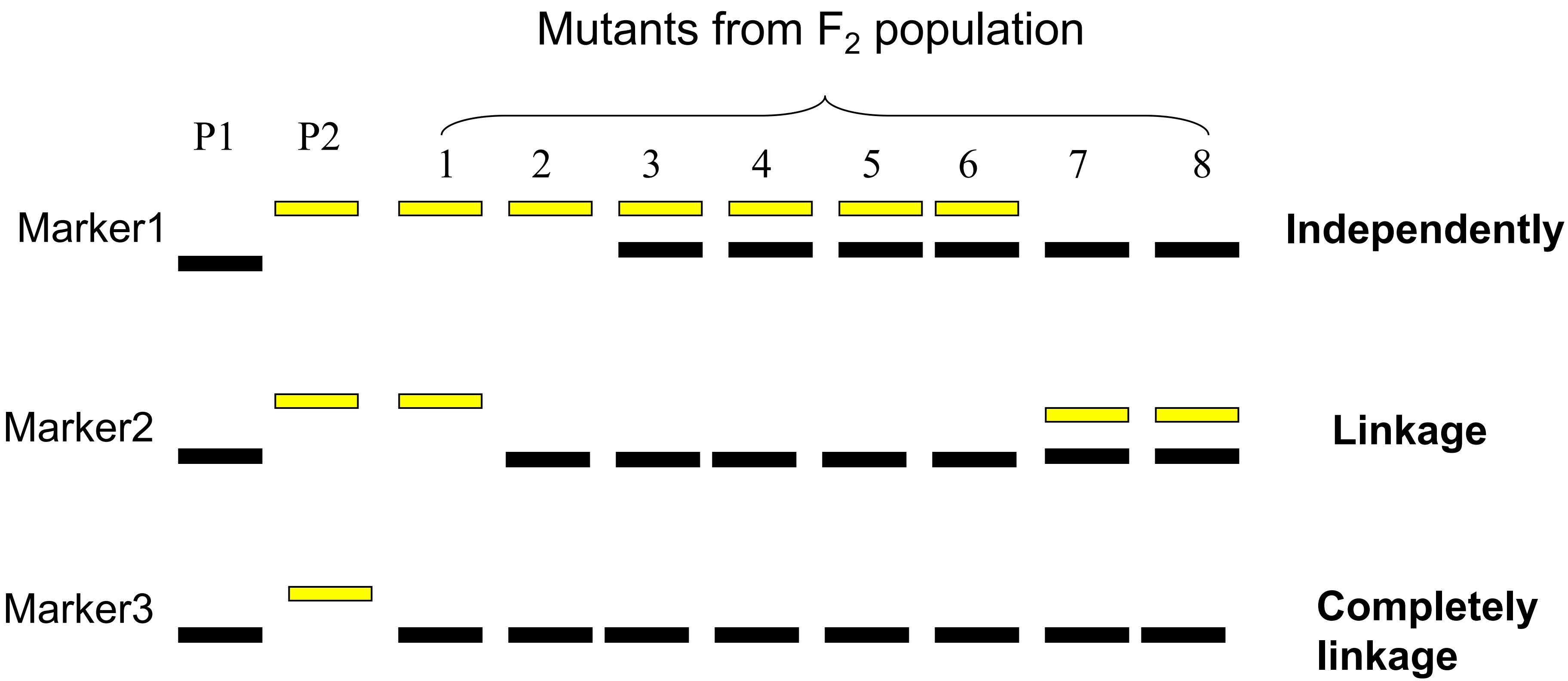
Independently r= 0.5

recombination frequency: r =发生交换的配子数/总配子数

Map based cloning principle



Map based cloning principle



RFLP, Restriction frangment length polymorphisms

Molecular marker: RFLPs (1980, Davis Bostein et al.)

Polymorphism: a genetic locus has different forms, or alleles

Digesting DNA from two individuals with Restriction endonucleases may yield fragments of different lengths.

Restriction endonucleases: are enzyme which cut DNA into defined fragments.

RFLP, Restriction frangment length polymorphisms

Recognition seq: Restriction endonucleases cleave DNA symmetrically in both strands at short palindromic (symmetrical) recognition seq to leave a 5’-phosphate and a 3’-OH. They leave blunt ends, or protruding 5’- or 3’-termini.

5’-GAATTC-3’

3’-CTTAAG-5’

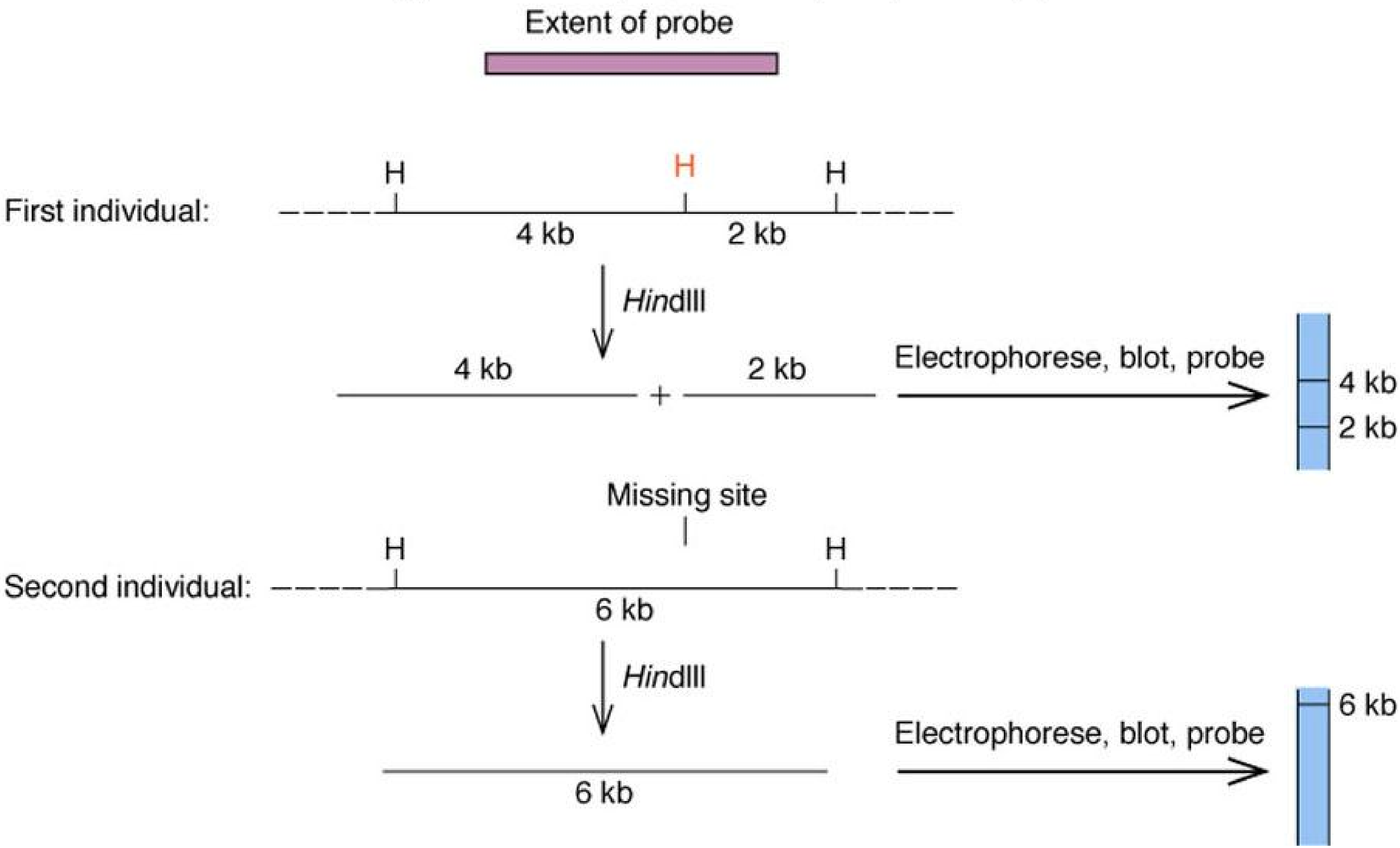
EcoRI

5‘-CCCGGG-3’

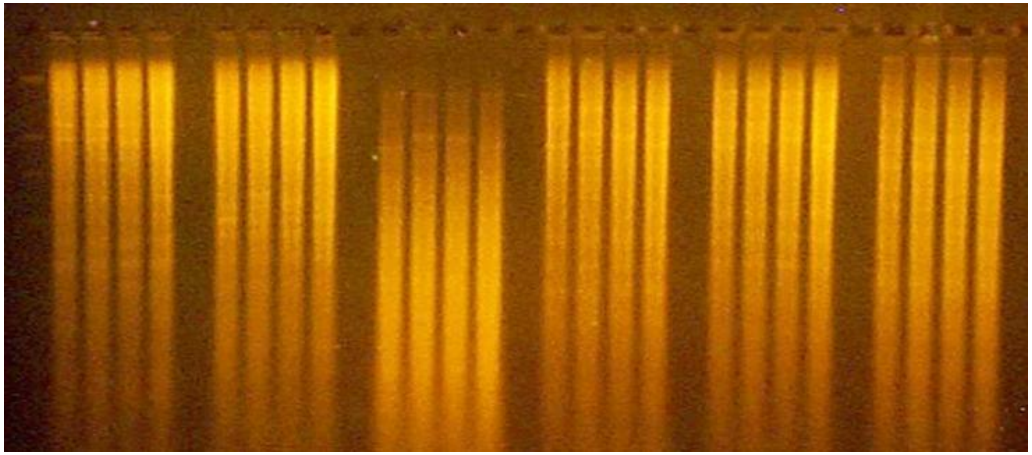
3’-GGGCCC-5’

SmaI

A 6 bp recognition sequences will occur on average every $4^6 = 4096$ bp in random DNA sequences.

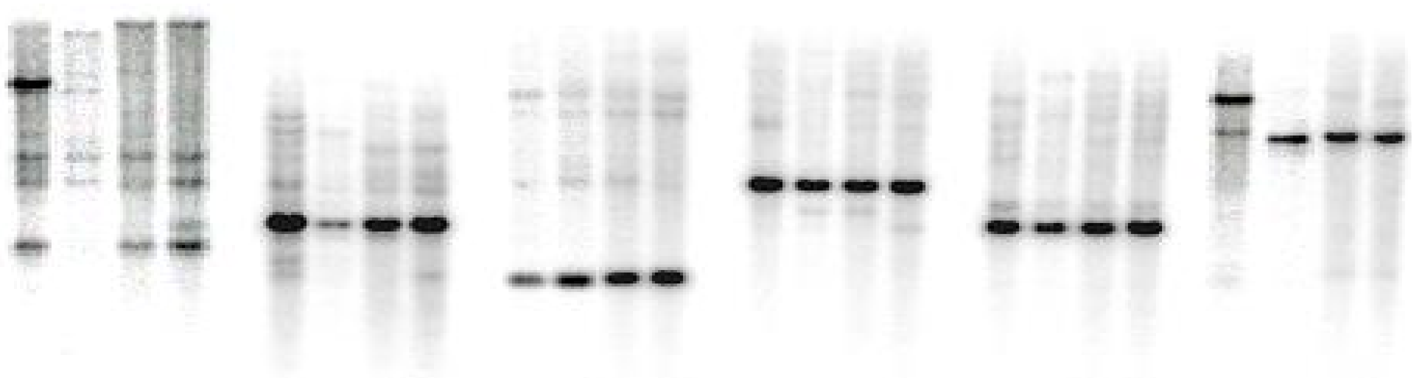

Extent of probe
First individual:
H
H
H
4 kb
2 kb
HindIII
4 kb
2 kb
Electrophorese, blot, probe
4 kb
2 kb
Missing site
Second individual:
H
H
6 kb
HindIII
6 kb
Electrophorese, blot, probe
6 kb

RFLP southern hybridization



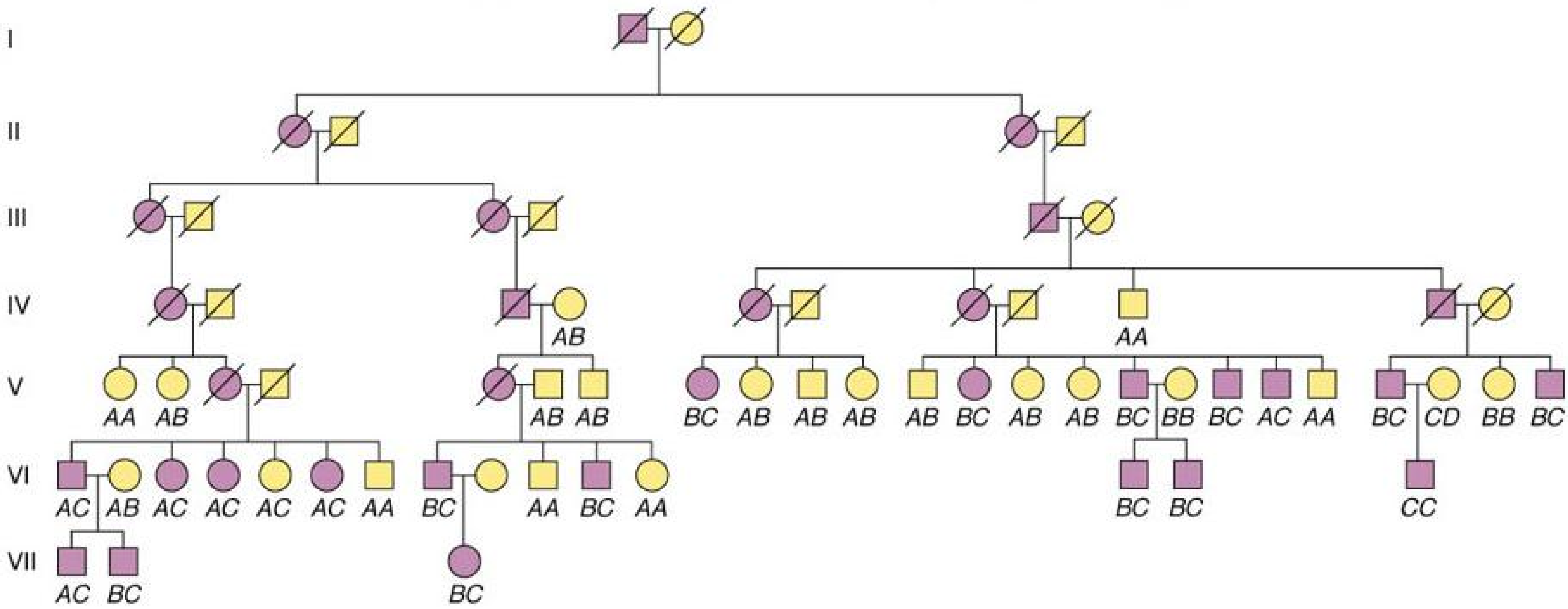
DNA digestion and electrophoresis

BamHI DraI EcoRI EcoRV HindIII XbaI



Nipponbare, Kasalath, IR20, 55178

An heroic story of identifying Huntington disease gene by positional cloning before genomic era



By using RFLP, Huntington disease are found be linked in 500 kb region in chromosome 4

CAG repeat Normal <35, mutant >35

After genomic sequence era, We can search gene very conveniently in database

Improve ORF (open reading frame) search program

- Codon bias密码子偏爱
- Exon-intron boundary
- Upstream control sequence. For example: CpG island in Vertebrate genomes

4 PCR based markers:
SSLP, STS, CAPS， dCAPS

Introduction of 4 popular markers for positional cloning after finish genomic sequence project

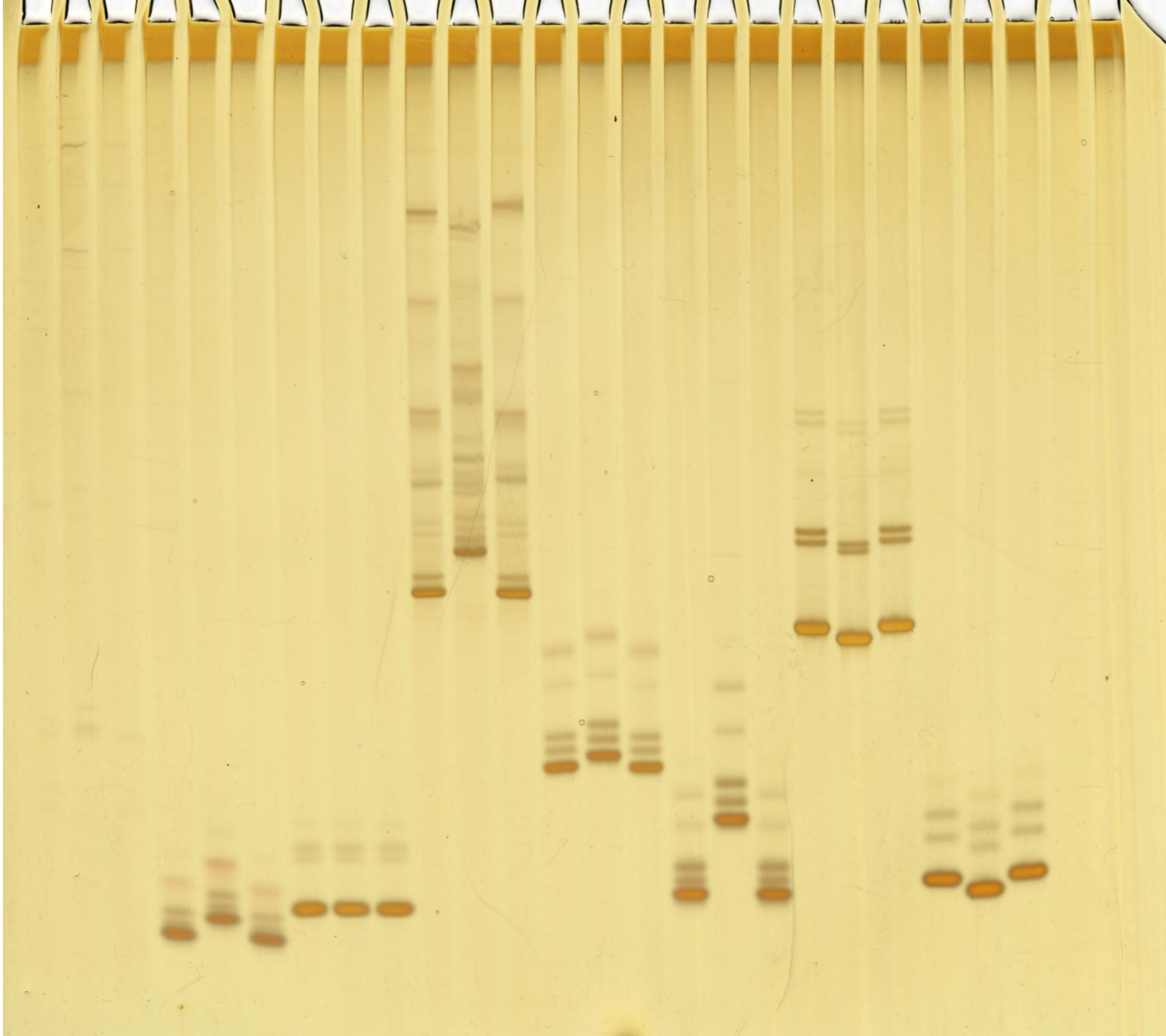
SSLP: Simple Sequence Length Polymorphism (=SSR, Simple sequence repeat)

SSLP: Microsatellite: 2-4bp long core sequence repeated over and over many times in a row. Highly polymorphic, widespread and relatively uniformly distributed, ideal markers for both linkage and physical mapping

P1: TCTATGAGAGAGGC

P2: TCTATGAGAGAGAGAGAGGC

P1 P2 Het



STS: Sequence Tagged Site

STS : Any sequenced DNA fragment, its specific position on chromosome is known. 序列标签位点

If any STS has polymorphism between alleles, and the polymorphism can be detected by PCR, then it can be a useful marker in linkage map

CAPS:

Cleaved Amplified Polymorphic Sequence

Principle:

1. By using the primers to get a PCR product (a specific DNA band)
2. Restriction digesting the PCR product using a specific restriction enzyme to detect polymorphisms of the digest products on agrose gel
3. Genotyping the polymorphic bands among the population

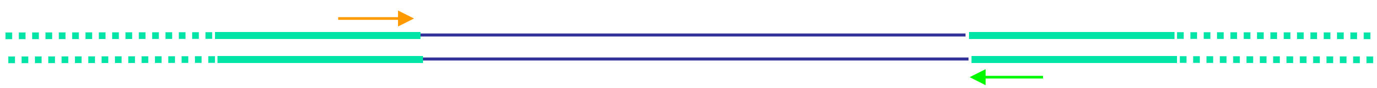
CAPS:

Cleaved Amplified Polymorphic Sequence

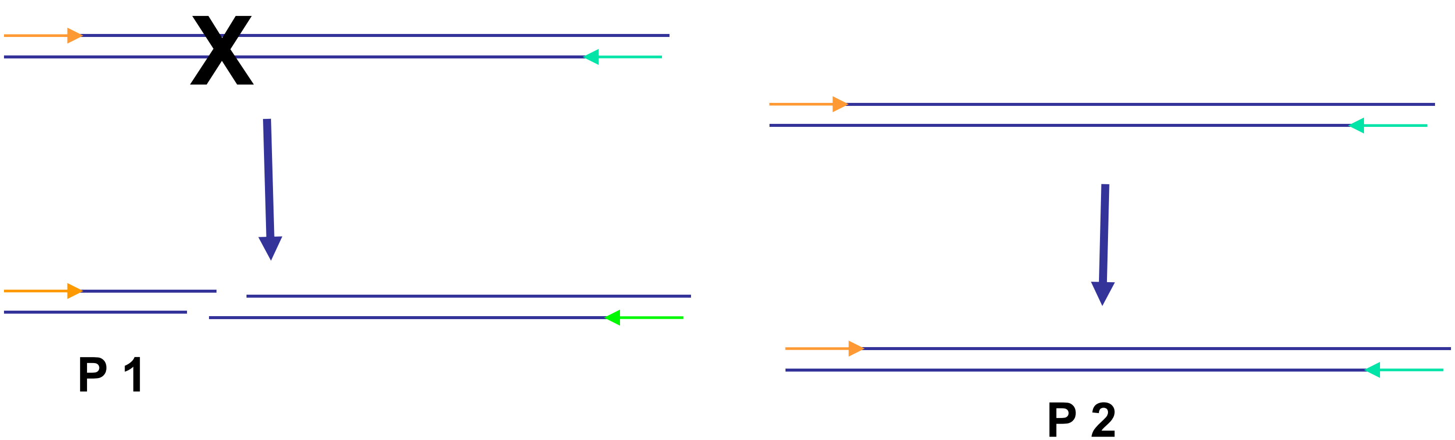
Process:

1. Amplify Target Sequence

Design primers ( ) complementary to flanking regions



2. Cut with a restriction enzyme that differentiates alleles



CAPS:

Cleaved Amplified Polymorphic Sequence

Process:

3. Alleles can be differentiated by size based on loss or gain of restriction site; May be able to analyze on agarose gel

M P1 P2 Het

dCAPS: Derived Cleaved Amplified Polymorphic Sequence

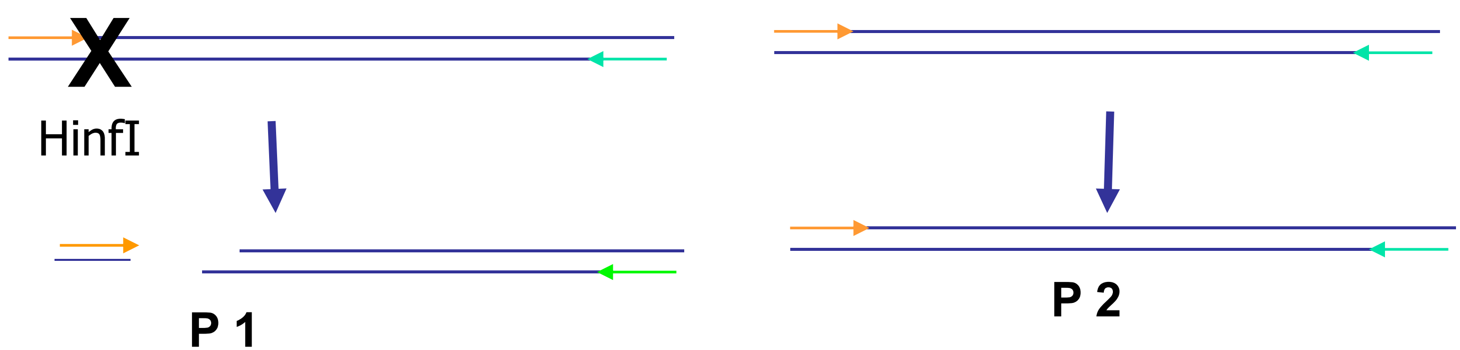
SNP

P1 ------------------GCTTCCACAAGTC-------------------------------------

P2 ------------------GCTTCCACAAGTT-------------------------------------

Design a forward primer: -----GCTTCCACGAGT-

HinfⅠ recognize site: GANTC



Each SNP can be designed as a dCAPS marker

Procedure of map based cloning

1. F_2 population construction. Cross the rice mutant with another Genotype rice to get F1, self pollinate of F1 to get F2 progeny for maping.

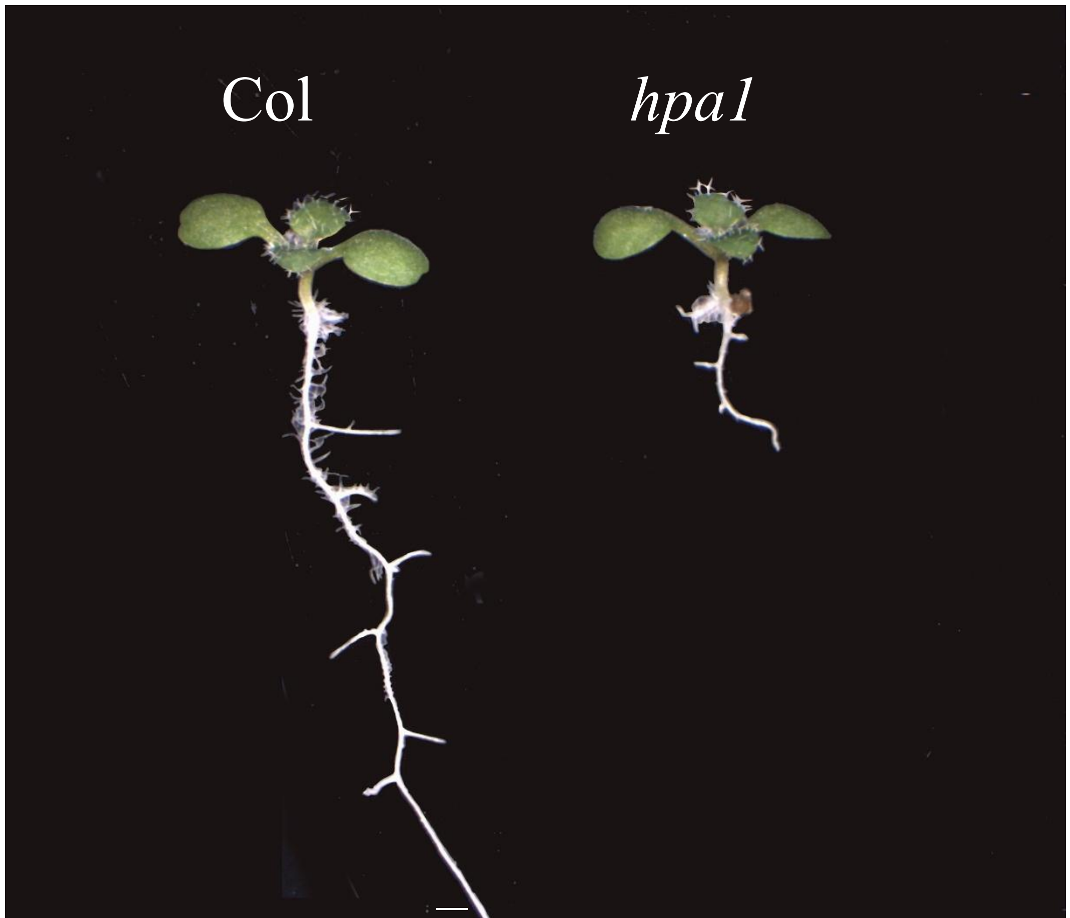
2. Initial mapping(初步定位). Screen 30 mutant phenotype plants in F2 progeny. Extract DNA of these 30 mutants. Detect the polymorphic bands among this small population using markers on different chromosome. The completely linkage marker are find

3. Fine mapping(精细定位). Screen1000-2000 mutants in amplified F2 population. Extract DNA of these 1000-2000 mutants by pooling. Detect the polymorphic bands among this big population using markers which around the linkage marker which find in initial mapping. Finally a candidate gene contained region can be determined between 2 most linked markers.

4. Sequence the DNA in the region which determined in fine mapping to find candidate gene.

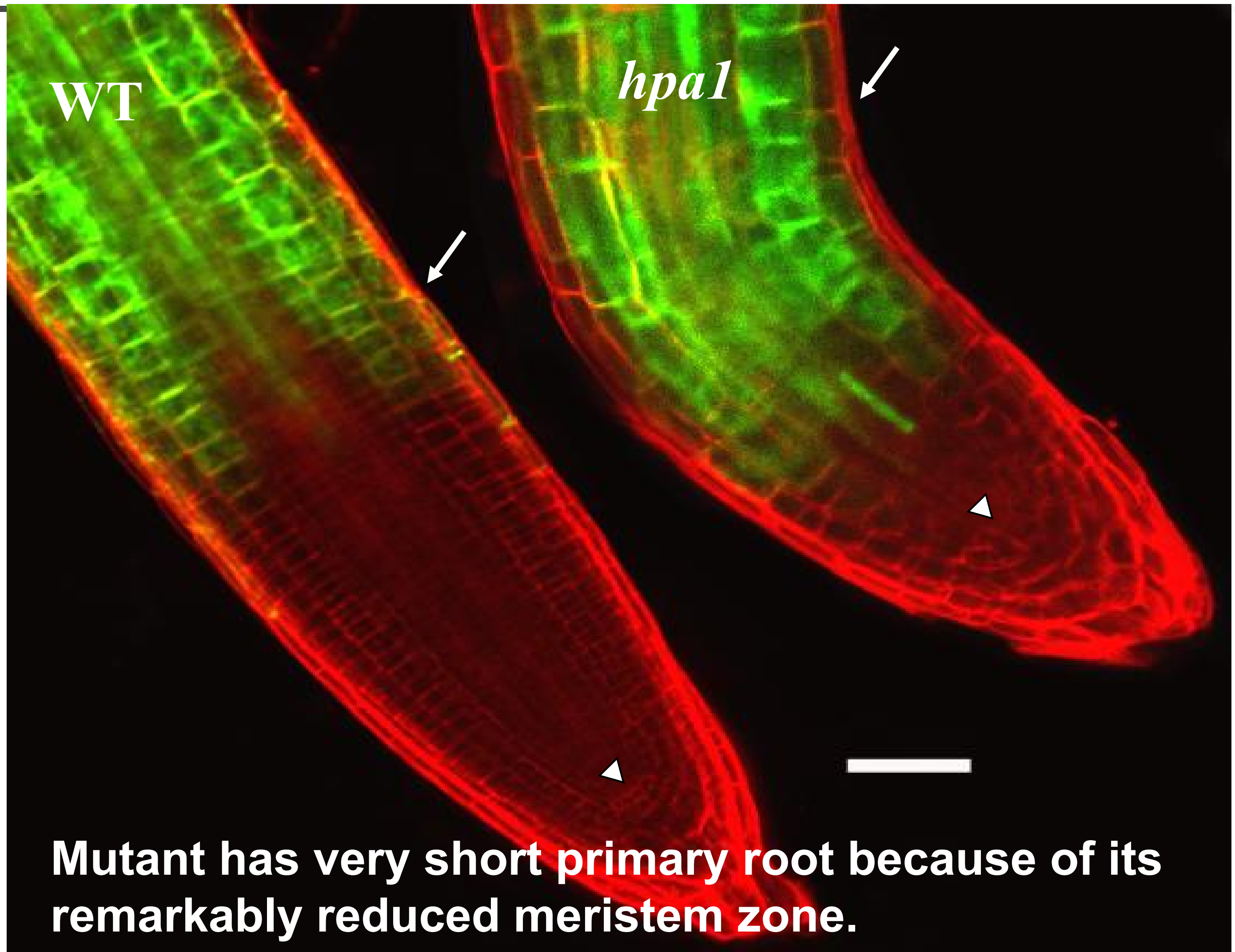
5. Transform wild type candidate gene into mutant to confirm this candidate gene function or mutant phenotype reoccur by RNAi or mutation (gene editing)

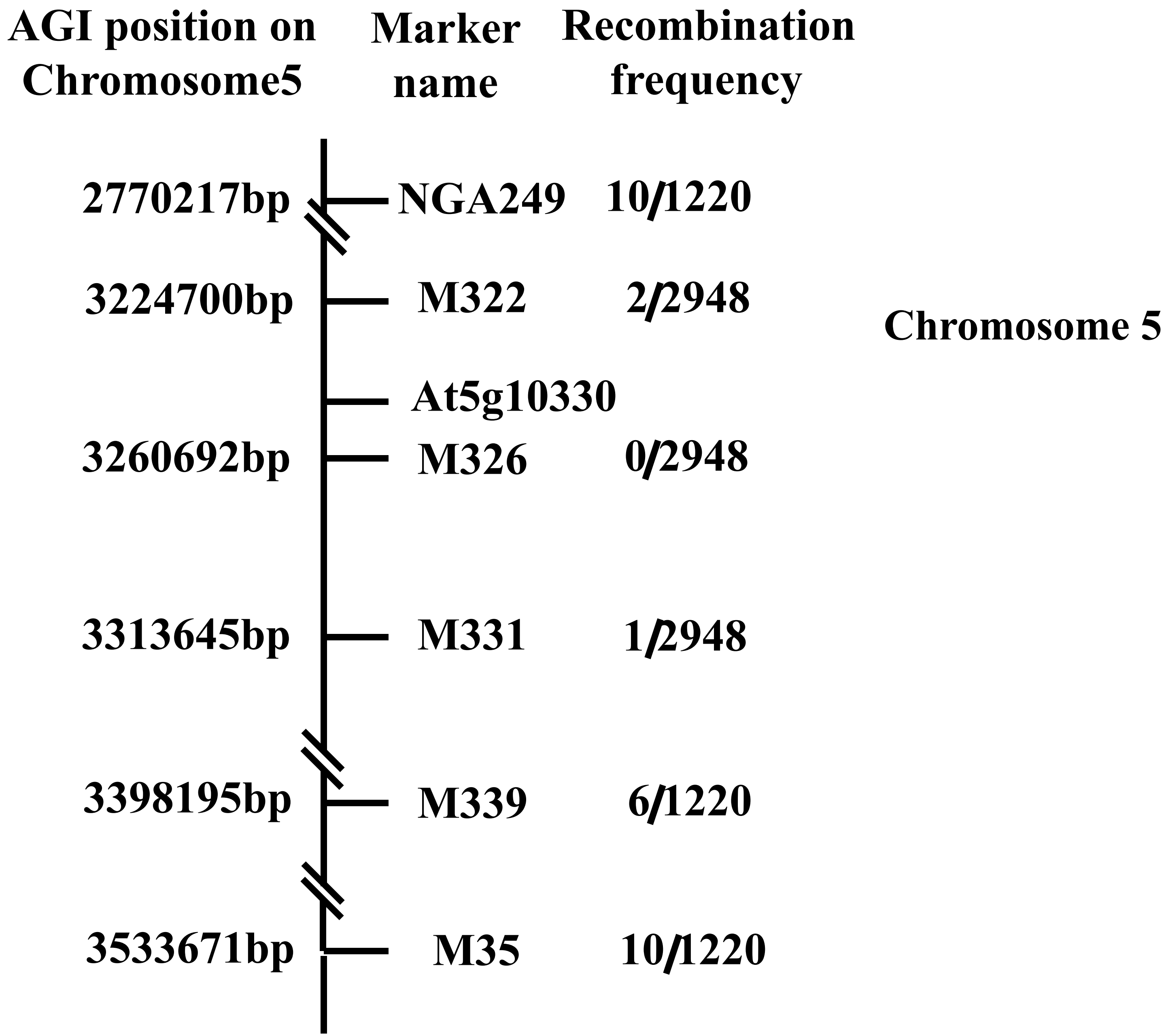
Lab story on mapping



We screened a short primary root mutant (plant physiology, 2006, Vol. 141 pp. 1425—1435)

Lab story on mapping



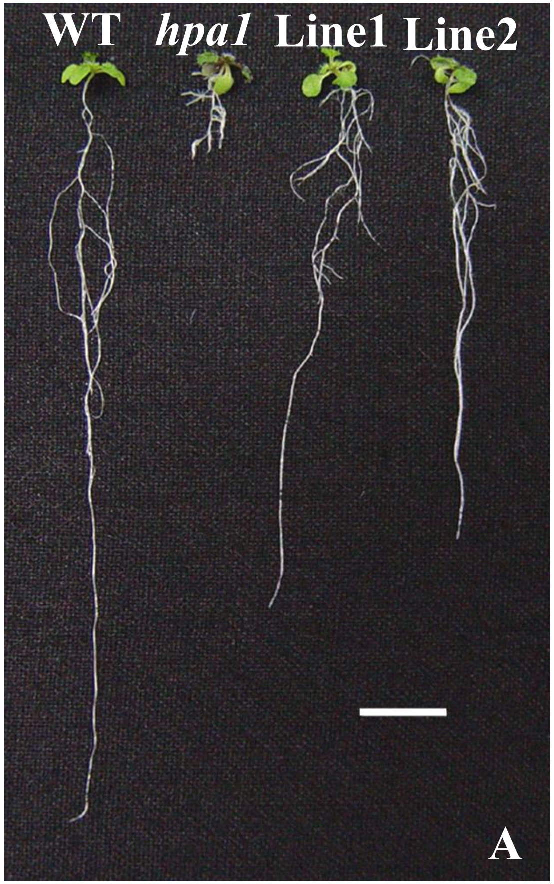


Fine mapping results with 2948 mutants from F2 population

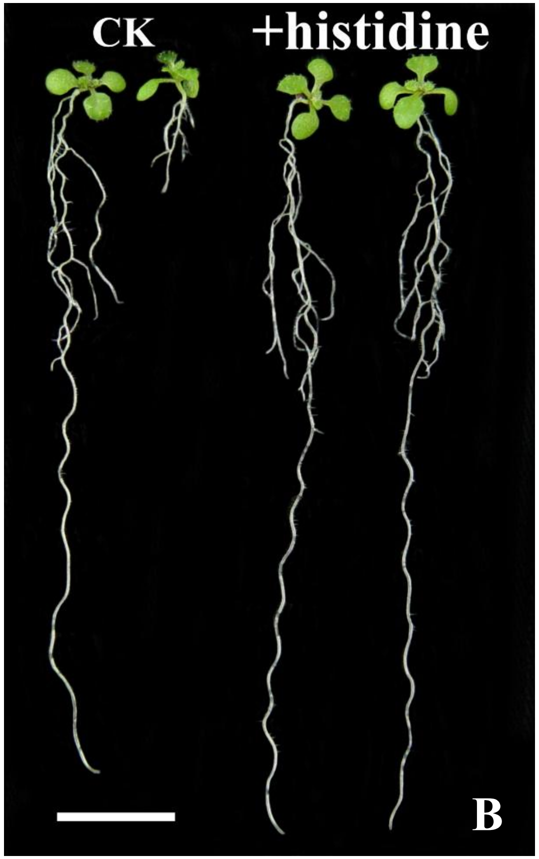
Lab story on mapping

```
                                    T
At5g10330   : DSFIRPHLRQLAAYQPILPFEVLSAQLGRKPEDIVKLDAN : 97
At1g71920   : DSFIRPHLRQLAAYQPILPFEVLSAQLGRKPEDIVKLDAN : 97
TobaccoHPA  : DAFIRPHLLKLSPYQPILPFEVLSTRLGRKPEDIVKLDAN : 93
riceHPA     : ASFIREHLRSLAPYQPILPFEVLSARLGRKPEDIIKLDAN : 66
AlgaHPA     : ASFVRPHLLALAPYKPIEPLDVLSARLGRPASEIVKLDAN : 69
YeastHPA    : KRIVRPKIYNLEPYR----------CARDDFTEGILLDAN : 35
E.coliHPA   : TDLARENVRNLTPYQS---------ARRLGGNGDVWLNAN : 37
```

Lab story on mapping



A: two independently transgenic lines with wild type *HPA* into mutant



B: Rescuing of the *hpa1* root growth defect by Histidine supplementation.

Gene cloning by using deep sequencing

- Create a segregation population
- Bulk the mutant phenotype lines and the wildtype lines and sequencing.

Genome-wide Association Study (GWAS) 分析

通过高通量分析比较病例与对照人群的遗传变异（SNP），基于**连锁不平衡（LD）**原理，将基因型与表型关联，从而定位影响复杂性状或疾病的易感基因位点。

01
样本收集
严格收集大样本量的病例与健康对照

02
基因分型
检测全基因组数百万个SNP标记位点，获取高密度基因型数据

03
统计分析
计算等位基因频率差异，筛选位点

04
结果验证
可视化结果，并在独立人群样本中重复验证

案例：人类身高的遗传解析

2007

利用GWAS研究，成功定位了第一个与人类身高显著相关的基因位点*HMGA2*。

2022

基于**540万+**样本的元分析，鉴定出超过**12,000**个独立遗传变异，累计解释了约40%的身高表型遗传度。

Student’s presentation (today)

Paper: LYCHOS is a human hybrid of a plant-like PIN transporter and a GPCR.Nature.2024,634:1238-

Presenter: Group 12